EVALUATION OF COWPEA, *VIGNA UNGUICULATA* L. WALP., FOR RESISTANCE TO THE COWPEA CURCULIO, *CHALCODERMUS AENEUS* BOHEMAN

by

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(Under the Direction of Peggy Ozias-Akins)

ABSTRACT

Cowpea, *Vigna unguiculata*, L. Walp, is a globally important food and forage legume. In the southeastern United States, the primary pest affecting cowpea production is the cowpea curculio, *Chalcodermus aeneus* Boheman. Current control strategies are insufficient, emphasizing the need for resistant germplasm. This study evaluated cowpea germplasm for resistance to the cowpea curculio and identified genotypes with potential resistance. Fourteen cowpea genotypes and a susceptible check were evaluated in four replicated, open choice field experiments over three years. A subset of five genotypes and a susceptible check were subsequently evaluated in a replicated no choice experiment. The findings from this research offered insights into the genotypes with established resistance and document additional resistance observed in the genotypes, which could assist in development of cultivars resistant to the cowpea curculio. This research is important for cowpea improvement and agricultural sustainability in the regions most affected by this destructive insect pest.

INDEX WORDS: Cowpea curculio, *Chalcodermus aeneus* Boheman, host plant resistance, *Vigna unguiculata* L. Walp.

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DEDICATION

I dedicate this work to my family and friends. To my partner, Blake, who has been my rock and kept me together throughout this journey. To my parents, Alan and Michelle, and my stepmother, Jen, thank you for always encouraging me to follow my passions. To my grandparents, Carol and Dick, for their support in my education and professional growth. To my sisters, Erin and Sara, thank you for being wonderful role models and for showing me that life has multiple paths to follow. You have shown me that there is no single way to achieve happiness and success. To my amazing nieces, Gabrielle and Madelyn, your curiosity, laughter, and endless energy inspire me every day. I hope you always keep that spark and never stop exploring the world. And to all my friends who have supported, I am thankful for your friendship.

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CHAPTER 1

INTRODUCTION AND LITERATURE REVIEW

Introduction

Cowpea, *Vigna unguiculata* L. Walp., commonly referred to as the black-eyed pea or southern pea, is one of the most important legume crops worldwide. Cowpeas are primarily cultivated in arid regions of the tropics, including areas of Asia, Africa, Southern Europe, the United States, and Central and South America (Singh, 2005; Timko et al., 2007). This multifunctional crop not only provides food for humans and livestock but also serves as a reliable income source for farmers and grain traders (Timko & Singh, 2008). Highly versatile and adaptable to diverse environmental conditions, cowpeas grow well in high temperatures and drought-prone areas, which has facilitated its widespread cultivation. This crop is a valuable component of cropping systems where soil fertility is limited. As a legume, it has a symbiotic relationship with nitrogen-fixing bacteria that enhances soil fertility, providing essential benefits for crop rotation and intercropping systems, particularly for resource-poor farmers (Nounagnon et al., 2024; Timko & Singh, 2008). Nearly all the above-ground plant material is consumed in various forms, including edible shoots, young leaves, immature pods, as well as green and dried seeds, which can also be processed into flour or paste. It has become an important crop in the local cropping and food systems of over 100 countries (Singh, 2020).

Cowpeas were domesticated in regions of East and West Africa over several thousand years ago and have spread across the world to almost every continent except Antarctica (Pan et al., 2023). Cowpea is a diploid species with 2n=22 chromosomes and is highly self-pollinating. The cowpea

genome is approximately 620 million base pairs, the smallest among legumes and one of the smaller plant genomes overall (Boukar et al., 2019). Cowpea belongs to the family Fabaceae, subfamily Faboideae, tribe Phaseoleae, subtribe Phaseolinae, and genus *Vigna.* The cultivated cowpea, *Vigna unguiculata* subspecies *unguiculata,* has been subdivided into five cultivar groups: *unguiculata, sesquipedalis, textilis, biflora,* and *melanophthalmus,* which are also recognized as five subspecies. The two primary cultivated cultivar groups are subspecies *unguiculata* and subspecies *sesquipedalis* (Boukar et al., 2019).

In recent years, there has been increased interest in legumes like cowpeas due to global challenges in food security and nutrition in response to a growing population. Low-input crops such as cowpeas have improved the livelihoods of millions of subsistence farmers in Africa, Asia, and South America by providing food and income security (Gonçalves et al., 2016). Cowpeas provide dietary protein, nutrients, and energy that are essential for enhancing the health and well-being in many regions of the tropics, establishing it as an important marketable crop for developing countries (Mahesha et al., 2022). Major world producers of cowpeas include Nigeria, Niger, Burkina Faso, Ghana, Mali, and Sudan. Brazil is also a major producer; however, it does not separate its cowpea production quantity from that of other grain legumes. The rise in global production to 9.7 million metric tons is likely due to cowpeas' numerous physical and nutritional characteristics. In contrast, production in the United States has declined to 12,927 metric tons in 2023 (FAOSTAT, 2023). This is presumably due to increased pest pressure in the main production areas of the southern states.

Historically, cowpeas were primarily grown as forage for cattle in the United States until World War II. Recently, cultivation practices have shifted towards a larger scale for various edible forms, including processed vegetables, dry beans, and market garden crops. Different varieties of

cowpeas have also gained popularity in home gardens across the southern states, from Texas to the Carolinas and Virginia. While Georgia, Arkansas, and Tennessee are the leading producers of processing peas, Texas and California are the leading producers of dry peas (NASS, 2021). Production in the United States peaked in 1937 and has drastically declined in the southeastern region since that time; however, production of dry black-eyed peas in California has continued to rise (Hall et al., 1997). The cowpea curculio, *Chalcodermus aeneus* Boheman, the main insect pest affecting cowpeas in southern states, remains unreported in California. As global challenges in food security continue to rise, the multifunctional nature of cowpea makes it a key commodity in future agriculture strategies, benefiting not only the environment but also the communities that rely on it.

Literature Review

Plant growth and morphology

Among the cultivated crops, cowpea has the most variability in growth patterns, morphology, maturity, and seed characteristics. Various environmental factors, including photoperiod and climatic conditions such as temperature and rainfall, can substantially influence both plant height and morphological traits (Singh, 2020). As a warm-season annual, cowpea requires a maturation period ranging from 50 to 160 days to progress from seed to seed.

The cultivated genotypes of cowpea are diverse in growth habit, leaf morphology, flower color, pod characteristics, pod color, seed type, and seed color. Growth habits may be categorized as prostrate, semi-prostrate, semi-erect, erect, or climbing, with growth patterns classified as either determinate or indeterminate. Leaf morphology is divided into four types: sub-globose, sub-hastate, globose, and hastate/lanceolate (Carvalho et al., 2017; Singh, 2020). Pods show variability in size, shape, and color, which can be cylindrical, straight, or curved, with mature colors ranging

from straw to brown or dark purple. Seeds can be either round or kidney-shaped, with textures that may be smooth, rough, or wrinkled. Their colors can include white, cream, green, red, brown, or black, and may be solid, speckled, or patterned (Singh, 2020; Timko & Singh, 2008).

The flowers of cowpea display a range of colors and possess six documented patterns of pigmentation. The flowers also have extra-floral nectaries that attract beneficial insects (Boukar, Togola, et al., 2019). The flowers have a brief life cycle, typically opening in the early morning and closing around midday, after which they generally undergo wilting and collapse (Kuo & Pate, 1985). Photoperiod does not have much impact on leaf appearance or other vegetative growth aspects; however, it does strongly affect reproductive development. Different cowpea genotypes have varying sensitivity to photoperiod, influencing the initiation and development of floral buds. Certain genotypes are classified as day-neutral, while others respond to a broader range of photoperiods (Craufurd et al., 1997). A number of cowpea genotypes display varying degrees of sensitivity to photoperiod, which can lead to delayed flowering, as well as sensitivity to temperature (Ehlers & Hall, 1996). Elevated temperatures generally promote accelerated flowering in both photoperiod sensitive and insensitive genotypes (Singh et al., 2023).

Germplasm and genetic diversity

Germplasm collection, characterization, evaluation, and conservation are fundamental components of cowpea research and breeding programs. These processes are important for understanding the genetic diversity of cowpea accessions, identifying desirable agronomic traits, and developing effective conservation strategies (Boukar et al., 2019). Cowpea germplasm collections are preserved at several major institutions: the International Institute of Tropical Agriculture (IITA) in Ibadan, Nigeria, which maintains over 17,000 cowpea accessions; the USDA Plant Genetic Resources Conservation Unit (PGRCU) in Griffin, Georgia, USA, with more than

8,000 accessions; and the University of California, Riverside (UCR), USA, which contains over 4,000 accessions. These gene banks are the largest repositories of cowpea biodiversity. Additionally, the World Vegetable Center and several national gene banks maintain large collections of regionally and locally adapted types. The accessions maintained in these gene banks exhibit important phenotypic variation in qualitative traits such as plant type, seed coat color, and flower color, as well as in quantitative agronomic traits including yield, maturity, and stress tolerance (Boukar et al., 2019). It is believed that cowpeas evolved from a limited section of wild germplasm from the species *Vigna unguiculata* and closely related species, with only partial gene flow occurring between cultivated and wild types. Despite the existence of natural hybrids between wild and cultivated cowpea genotypes, as well as wide phenotypic variation among the accessions, a narrow genetic diversity is typically observed within the cultivated gene pool (Ehlers & Hall, 1997). This limited genetic diversity is largely attributed to the self-pollinating nature of the crop. Most of the genetic diversity is found in the wild genotypes, making wild germplasm an essential resource for breeding programs (Nounagnon et al., 2024).

Recent research on the diversity of cowpea germplasm has focused on the analysis of molecular markers. All types are used in the diversity analysis; however, Single Nucleotide Polymorphisms (SNPs) have been described as more effective than other markers for analyzing population structures within cowpea germplasm due to their genomic abundance (Xiong et al., 2016). Several molecular studies utilizing the USDA cowpea germplasm collection have been successful in their research findings. One study using KASP (Kompetitive Allele Specific PCR) SNP markers identified greater genetic variation across geographical regions and subspecies, finding that germplasm originating in different regions exhibits abundant gene flow (Potts et al., 2024). Additionally, a Genome-Wide Association Study (GWAS) that analyzed 373 cowpea

accessions linked SNPs to nine very important traits such as pod length, seed antioxidant content, plant maturity, flower color, and growth habit (Wu et al., 2024). These studies highlight the importance of germplasm collections and genetic diversity in cowpea breeding and the potential for crop improvement through molecular techniques.

Recent research on cowpea improvement has applied the use of transgenic technology to develop the first genetically modified (GM) *Bt* cowpea. Development of the *Bt* cowpea was initiated to provide resistance to the bean pod borer, *Maruca vitrata* (Fabricius), the most damaging pest to cowpea in Sub-Saharan Africa. Genetic engineering was implemented after researchers faced challenges in successfully crossing cowpea with wild relatives with resistance to this insect pest (Boukar et al., 2020). The GM cowpea variety carries an insecticidal *Cry1Ab* gene encoding a *Bacillus thuringiensis (Bt)* protein which is toxic to certain insects (Mahesha et al., 2022). Nigeria is the first country to approve cultivation of the first GM insect resistant cowpea, SAMPEA 20-T, Pod Borer Resistant (PBR) Cowpea in the world (Boukar et al., 2020).

Cowpea market preferences and breeding goals

In the United States, cowpea cultivars are grown for horticultural purposes, processing, and fresh market production (Singh et al., 2023). Processing varieties are used for canning, freezing, or producing dry peas (Ehlers et al., 2002). Cowpeas are divided into three primary types: black-eyed, cream, and crowder, each grown for different purposes. The black-eyed type is the most widely grown, known for high yield, and is primarily used in the dry pea industry. Pink-eyed is a variation of the black-eyed type and is favored in home gardens, fresh markets, and for processing. Crowder types, known for their larger peas, are often used for canning (Fery, 1990). Cream peas are a preferred table vegetable for American consumers and are important for both home gardens and processing. Cultivars with purple hulls are particularly favored in the fresh market due to their

extended shelf life and popularity among home gardeners (Ehlers et al., 2002; Hall et al., 1997). A recent development in these pea types is the incorporation of a persistent green seed coat color in dry seeds. These varieties are valuable for the frozen food industry and fresh market because they closely resemble fresh shelled cowpeas and reduce harvesting costs by allowing mature pods to be harvested, cleaned, and stored dry (Hall et al., 1997).

Commercial and fresh market production has specific standards for cowpea cultivars. Processing varieties should have uniform maturation, with 85-90% of the pods reaching maturity simultaneously, along with an upright and compact growth habit. Early maturity is also important. Fresh market cultivars should yield attractive, well filled pods that are easy to shell by hand. These pods are required to have an extended harvestable period, regardless of whether they mature uniformly. Varieties that dry quickly in the field or have a short shelf life are unsuitable for fresh market purposes (Fery, 1990).

Cowpea breeding has been a challenging task for scientists due to both abiotic and biotic constraints, and the fact that cowpea variety requirements vary from region to region based on seed color preference, use patterns, maturity, and growth habit. Breeding programs prioritize target production zones, considering cropping systems, consumption preferences, and major production constraints specific to their areas (Singh et al., 2023). No single cowpea variety can be considered suitable for all countries and regions. Therefore, efforts concentrate on developing varieties for specific regions. However, the primary objective for all cowpea breeding programs is to achieve higher grain yield and improved grain quality. Additionally, breeders aim to incorporate a diverse range of abiotic and biotic stress resistance traits into their breeding efforts (Singh et al., 2023; Singh, 2020).

Insect pests of cowpea

The productivity of cowpea is affected by various biotic stresses, including diseases and insect pests. Numerous insect pests of cowpea are known throughout the world with at least one major pest causing losses at every growth stage. Key insect pests include aphids, thrips, the Maruca pod borer, a complex of pod-sucking bugs, and the cowpea weevil. In addition, certain location specific pests exist, such as the bean fly in Asia and East Africa and ootheca beetles in the wetter regions of the tropics (Timko & Singh, 2008). The losses from these insects can be devastating, ranging from 50% to 100% (Togola et al., 2017).

In the state of Georgia, several insect pests affect cowpeas, including thrips, aphids, lygus bugs, stink bugs, the cowpea weevil, and the cowpea curculio (Osipitan et al., 2021). The key pest of the southeastern United States is the cowpea curculio. During the adult phase of its life cycle, the cowpea curculio crawls onto the plant and feeds on the leaves, flower buds, and developing pods. It typically consumes leaves and stems until pods become available. Green pods that are half grown to maturity, are preferred (Arant, 1938).

Both male and female adult cowpea curculios will cause damage by feeding on the external surfaces of the pods. Following this feeding, the female deposits an egg through the pod wall into the pea inside the pod, through a process called oviposition. The female cowpea curculio can lay nearly 300 eggs within a span of 45 days (Arant, 1938; Capinera, 2001). Generally, one egg will be deposited per puncture. These eggs are oval, white, and measure less than 1 millimeter in length (Capinera, 2001). The duration of the egg stage ranges from three to six days. Larval development is influenced by temperature, with the four larval instars typically taking between six and nine days to complete (Capinera, 2001). The larvae may remain within the pod, feeding and developing for up to 27 days (Arant, 1938). The primary source of damage to the crop occurs during the larval

stage, as the larvae feed directly on the developing pea. However, because the larvae are protected by the pod, they are often not targeted by insecticidal control measures (N'Guessan & Chalfant, 1990).

The soil phase of the cowpea curculio's life cycle begins when the larvae emerge from the pod. They create an exit hole in the pod wall, fall to the ground, and burrow down to approximately three inches, where they form a pupal chamber and pupate for up to 19 days (Arant, 1938; Capinera, 2001). The larvae and pupae are light yellow, and the pupae closely resemble the adult stage (Capinera, 2001). The developmental stages within the soil include prepupal, pupal, and resting adult stages. After transitioning to the adult stage, it has been observed that these insects will remain in the soil for one to sixteen days, but this likely refers to the spring-summer generation (Arant, 1938). Despite possessing fully developed wings, flight is rarely observed, and it is likely the cowpea curculio crawls to its host plant upon exiting the soil phase (Capinera, 2001).

Following a period of overwintering, cowpea curculios emerge in the spring coinciding with the initial planting of cowpeas. These adults, having starved during the winter months in the soil, begin feeding directly on the plants prior to fruiting. In addition to causing damage to the pods through both feeding and oviposition, adult cowpea curculios also damage foliage and flower buds. The complete life cycle of the cowpea curculio spans approximately 23 to 53 days (Arant, 1938). Typically, cowpea curculios produce two generations annually and exhibit three distinct peaks of activity: the first cycle occurs between mid-April and mid-June, the second cycle between late July and early October, and the third cycle from mid-November to mid-December (Riley et al., 2015).

Cowpea curculio management

Managing cowpea curculios in the field requires implementing various cultural practices, including maintaining fields free from any alternate host plants for this pest (Capinera, 2001).

While cowpeas are the preferred host, adult cowpea curculios will feed on or utilize other plant species before, during, and after the cowpea cropping season. Some alternative host plants include young cotton, string beans, lima beans, strawberries, snap peas, English peas, common vetch, sow thistle, evening primrose, and sickle pod. However, cowpeas are the main reproductive host of significance (Arant, 1938; Sudbrink et al., 1998). The absence of alternative hosts may lead to a decline in cowpea curculio populations for the rare flying pest. Additional cultural control practices include tillage, weed management, and field sanitation. Routine tillage and the destruction of host crops post-harvest can help reduce cowpea curculio populations by disrupting the soil phase of their life cycle, potentially decreasing emerging populations in the spring and eliminating alternative weed and crop hosts. Sanitation efforts focused on removing debris from the field post-harvest help to prevent the survival of overwintering cowpea curculios (Riley & Sparks, 2014).

The effectiveness of pyrethroid insecticides in controlling cowpea curculios has diminished, resulting in limited control options for growers (Riley & Sparks, 2019). Current recommendations primarily focus on foliar spray applications during the flowering and fruiting stages. However, these applications do not target the most damaging stage of the cowpea curculio life cycle, the larvae, which remain protected within the pods. While biological control methods involving natural enemies and fungi have been assessed, further research is necessary to provide reliable recommendations for cowpea growers (Schmidt et al., 2018).

Host plant resistance to insects

Host plant resistance provides an effective and sustainable approach for managing insect pests by minimizing both insect occurrence and severity of damage. Painter (1951) defined host plant resistance as "the relative amount of heritable qualities possessed by a plant that influence

the ultimate degree of damage done by an insect." This concept involves various plant traits that deter or reduce insect feeding and reproduction. Host plant resistance involves three complex mechanisms: antixenosis, antibiosis, and tolerance. These mechanisms are interrelated, and any one or a combination of the three may be present in the resistance of host plants to insect pests (Togola et al., 2017).

The term "antixenosis" was introduced by Kogan and Ortman (1978) to replace "non-preference," as it refers to the insect's response rather than the characteristics of the host plant itself. Antixenosis describes how specific attributes of a host plant influence insect behavior in ways that reduce the chance of an insect utilizing the plant for feeding, oviposition, or shelter. The main effect of this mechanism is characterized by physical and biochemical factors in the plant that repel or disrupt insect behavior (Togola et al., 2017).

Antibiosis describes the adverse effects that a host plant exerts on insect pests due to the presence of toxic substances or the lack of essential nutritional components. For plants exhibiting antibiosis, insects may experience negative impacts on growth, development, survival, and reduced fecundity (Painter, 1951). The primary effects of this mechanism involve chemical compounds, including secondary metabolites such as alkaloids, terpenes, and phenolics, which can disrupt feeding, development, and reproduction in insects (Togola et al., 2017). Recently, the effects of an α-amylase inhibitor, a resistance factor from the common bean (*Phaseolus vulgaris* L.), have been investigated in the resistance of αAI-1 transgenic cowpea dry grains to several species of bruchid beetles (Coleoptera: Chrysomelidae) and has been shown to reduce digestion (Higgins et al., 2013; Lüthi et al., 2013).

The third mechanism of resistance is tolerance. This mechanism does not involve the degree of insect utilization of the host plant or the amount of damage inflicted but rather lies in the

response of the host plant to a given level of damage inflicted by insect feeding and oviposition (Van Driesche & Murray, 2004). It is defined as the ability of a host plant to recover or compensate for insect damage in ways that alleviate or reduce the impact on yield or fitness. For plants in general, common factors involved in tolerance include increases in photosynthetic rates, high growth rates, and activation and changes in patterns of resource allocation after injury (Smith, 2005). In cowpeas, tolerance has been associated with some physiological factors such as plant growth habit, recovery capacity from pest injury, early maturity, and high flower production (Togola et al., 2017).

Resistance in cowpeas to the cowpea curculio

Research on resistance in cowpeas to the cowpea curculio started over 50 years ago. Initial screening trials indicated that various cowpea genotypes exhibited differing levels of susceptibility to the cowpea curculio; however, no single genotype demonstrated a high level of resistance. These findings suggested that the observed low to moderate levels of resistance stem from multiple resistance mechanisms (Cuthbert et al., 1974). The objective of subsequent research was to understand the mechanisms or factors contributing to the varying levels of resistance expressed by different genotypes to facilitate the development of new varieties resistant to the cowpea curculio. Most of the research focused on prominent genotypes cultivated in the United States and was used in efforts to breed new cultivars for the areas most affected by the cowpea curculio infestation.

One previously identified factor influencing resistance reactions is the physical characteristics of the pod wall. Characteristics such as the thickness or toughness of the pod wall have been shown to indicate an antixenosis response by serving as barriers to feeding and/or oviposition, thereby preventing penetration by adult female insects (Chambliss & Rymal, 1980; Fery & Cuthbert, 1975). Additionally, the presence of specific chemical compounds can function

as either feeding stimulants or deterrents, depending on the presence or absence, and quantities. For example, a lack of water-soluble feeding stimulants in varying amounts within seeds, pods, or other plant parts contributes to antixenosis in certain genotypes. While other constituents of pods, including sugars, starches, and total nitrogen, have been positively correlated with feeding activity (Canerday & Chalfant, 1969; Chalfant & Gaines, 1973). Potential feeding deterrents have also been identified, including variations in fatty acid profiles and increased levels of tannins in resistant genotypes (Rymal & Chambliss, 1976). The influence of volatile substances related to antixenosis has also been investigated, with findings indicating that susceptible genotypes possess volatile feeding attractants that draw insects to their pods. In contrast, the pods of resistant lines exhibited fewer volatile components that function as non-attractants (Chambliss & Rymal, 1982). However, the economic viability of these resistance factors remains to be thoroughly researched.

Cowpea breeding efforts to develop resistance against the cowpea curculio have been challenging, particularly in recent years due to limited research. There is little literature available on cowpea and its resistance to the cowpea curculio since the turn of the 21st century. To date, only four documented cowpea cultivars released in the United States have resistance to the cowpea curculio, using the variety, Ala.963.8, and its progeny, CR 17-1-13, CR 18-13-1, and CR 22-2-21 (Cuthbert, 1976). Those cowpea resistant cultivars are AUBe, Bettergreen, Bettergro, and Carolina Cream which have pod characteristics that either interfere with the ability of adult insects to damage the seeds through feeding and oviposition, or make them less attractive to adult cowpea curculios, leading to reduced seed damage (Cuthbert, 1976; Singh et al., 1997). Thus, the resistance mechanisms displayed by the four cultivars is antixenosis reflected through a pod wall factor and/or additional factors affecting the cowpea curculio's behavior and utilization of it. The resistance to the cowpea curculio observed in the breeding material of the resistant cultivars was

documented to depend on various levels of antixenosis and antibiosis. Each of these mechanisms is genetically controlled and can be passed on to progeny, though selection must occur independently for each of the three resistance mechanisms (Cuthbert, 1976). While progress has been made, complete resistance has yet to be achieved.

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CHAPTER 2

EVALUATION OF COWPEA GENOTYPES FOR RESISTANCE TO THE COWPEA CURCULIO IN OPEN CHOICE EXPERIMENTS[1]

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Abstract

The cowpea curculio, *Chalcodermus aeneus* Boheman, is the primary insect pest affecting cowpeas, *Vigna unguiculata* L. Walp., in the Southeastern United States. Current control strategies involve foliar insecticide applications targeting the above-ground adult life stage, but these methods do not provide sufficient control. This study evaluates diverse cowpea germplasm for resistance and identifies genotypes with potential resistance to the cowpea curculio. Fourteen genotypes and a susceptible check were evaluated in four replicated, open choice field experiments over three years. Insect damage was recorded on sub-samples of mature fresh green pods, measuring the feeding and oviposition marks on pods and peas and the number of larvae remaining in pods. This research provides additional data on genotypes with known resistance reactions and documents genotypic responses in a growing region with heavy curculio pest pressure. Research results will assist in future development of cultivars resistant to the cowpea curculio.

Introduction

Cowpea, *Vigna unguiculata* L. Walp., also referred to as southern pea or black-eyed pea, is a significant crop in the United States and has played an important role in agriculture worldwide. In the southeastern region of the country, it is one of the most culturally significant specialty crops (Riley & Sparks, 2019). Historically in the United States, cowpea was planted for forage, soil management, and human consumption when production peaked at 345,000 metric tons in 1937 (Osipitan et al., 2021). Nearly all the production occurred in the Southeast at that time. In the most intensely cultivated areas of the Southeast, the impact of the cowpea curculio, on damage to pods and peas, has been so severe in recent decades that a substantial portion of commercial production for fresh frozen peas has relocated out of this region to avoid the losses of this one pest (Riley &

Sparks, 2019). Economically unacceptable losses are occurring, and management of the cowpea curculio is crucial for sustainable cowpea production in the Southern regions of the United States.

Additionally, worldwide interest in cowpeas has risen in the last few decades. Specifically, there is increased interest in the nutritional value and dietary benefits that support cowpeas as a promising crop for future food security (Omomowo & Babalola, 2021). This has likely driven the recent increase in global production of cowpeas from 1.3 million metric tons in 1981 to 9.7 million metric tons in 2023 (FAOSTAT, 2023). However, in the traditional cropping area of the United States, production has declined from 36,605 metric tons in 1997 to 12,927 metric tons in 2023, even though the commodity price has risen during this time (NASS, 1997; FAOSTAT, 2023). Most production losses occurred in the Southeastern United States, and the decline in cowpea production in this region is attributed, at least in part, to the inability to consistently control the cowpea curculio. In addition to damaging the pods through feeding and oviposition, adults will damage cowpea foliage; however, the primary damage occurs as the larvae develop within the pod and feed on the seed (Arant, 1938; Capinera, 2001).

Currently, cowpea curculio control options are limited. The recommended insecticides are no longer adequate for controlling this insect pest, which has left growers with a lack of chemical control options to manage the cowpea curculio (Riley & Sparks, 2019). Biological control measures such as using natural enemies like entomopathogenic nematodes have been evaluated against the cowpea curculio and have shown potential for control (Schmidt et al., 2018). However, there are no current recommendations for growers. The best option for control is to decrease cowpea curculio damage through resistant germplasm. Development of a resistant cultivar, which meets the grower and consumer preferences with the most desirable traits for production, is an ideal solution. An understanding of host plant resistance to cowpea curculio and the mechanisms

involved will aid in the development of a resistant cultivar. Cowpea genotypes exhibit different types of resistance mechanisms to feeding and ovipositing curculios. These mechanisms, tolerance antixenosis, and antibiosis may present singularly or in a combination, providing cowpeas with various levels of natural resistance.

The goal of this research is to investigate host plant resistance in cowpea and to identify cowpea germplasm with resistance to the cowpea curculio. The potential presence of resistance was studied through several open choice experiments. The open choice experiments were conducted through evaluations of insect damage on cowpea genotypes that were planted in fields naturally infested with cowpea curculios. The objective of this study was to confirm the previously reported resistance levels exhibited by the genotypes and provide additional information on the individual genotype response relative to cowpea curculio response and potential mechanisms of resistance.

Materials and Methods

Plant material

Fourteen cowpea genotypes and a susceptible check were evaluated for resistance to the cowpea curculio (Table 2.1). All materials were obtained from the USDA-ARS cowpea germplasm collection in Griffin, GA. The genotypes selected were compiled from previous research reports available in GRIN Global of cowpeas exhibiting different resistance reactions to the cowpea curculio (see Appendix A). Additionally, the genotypes exhibit a range of phenotypic characteristics, including variations in seed size, seed color, and even seed coat texture (Figure 2.1). The pods differ in length, shape, and color. The growth habit of the mature plant depends on genotype, growth temperature, and photoperiod of the growing environment, with each of the

selected genotypes being either erect, semi-erect, or prostrate. The period of pod development and maturity ranges from 60 to 120 days for the selected genotypes according to GRIN entries.

Experimental Design

The evaluation of genotypes was performed over four separate experiments during three years at the Lang-Rigdon Farm Hook Field at the University of Georgia, Tifton Campus in Tifton, Georgia, where fields have naturally high populations of cowpea curculios. Fields were closely located on the same farm but had slightly different field positions for each test. Each test was arranged in a randomized complete block design. Field plots for Tests 1, 2, and 3 were planted on plastic beds, and Test 4 was not, due to weather conditions delaying planting. Each plot was between 7.62 to 9.144 m in length with plants spaced 30 cm apart. Each entry was replicated three times. In 2022, two separate tests were conducted and designated Test 1 (Latitude 31.517638, Longitude -83.546140) and Test 2 (Latitude 31.517641, Longitude -83.545652). The materials for Test 1 were planted in the greenhouse on March 15th, 2022, and transplanted in the field on April 19th. The materials for Test 2 were direct seeded on July 15th, 2022. In 2023, one test was conducted and designated as Test 3 (Latitude 31.516141, Longitude -83.547925). The materials were direct seeded on April 27th. One test was conducted in 2024 and designated as Test 4 (Latitude 31.517638, Longitude -83.546140). Only a subset of nine resistant genotypes and one susceptible were included in this test (Table 2.2). The materials were direct seeded on May 30^{th,} 2024. Two additional replications of the susceptible genotype were planted two and four weeks post initial planting on June 12th and June 27th, 2024. Due to the variability of maturity among the genotypes and fluctuation in peak insect activity, the additional replications of the susceptible genotype were expected to provide a more accurate evaluation of insect pressure at different collection times.

Sampling and Evaluation Methods

Plots were entirely harvested as pods matured over a two-week period to maintain a consistent harvest across all replications. The two-week period of harvesting plots varied due to differing maturities of the genotypes. The mature pod stage harvested was the fresh market standard green pods. A random sample of 100 pods was separated from the total pods harvested in each plot and used for insect damage evaluations (Figure 2.2). The 100-pod sample was categorized as damaged (stung) and clean (blemish free) (Figure 2.3.). The pods were hand shelled to assess pea damage and examined for cowpea curculio larvae remaining in pods. The total weight of harvested pods (g), the total weight of the 100 pod sample (g), weight of clean pods per sample (g), weight of stung pods per sample (g), number of pods with stings per sample, weight of peas shelled from clean pods (g), weight of peas shelled from stung pods (g), number of damaged peas per 100 peas from shelled stung pods, and number of larvae per sample were recorded. The harvest dates (from planting or transplanting to harvest dates) were recorded. All data were analyzed using Proc GLM in SAS, and mean separation determined by LSD ($P<0.05$).

Results

This study evaluated insect damage on different genotypes in open choice experiments to assess insect host preference and overall resistance of 14 genotypes in a natural field environment. All 14 genotypes were evaluated in Test 1, 2 and 3. Later, a subset of nine genotypes were evaluated in Test 4 (Table 2.2). Genotypes that had no pod set in Test 1 were PI 582519, PI 598343, and PI 533010 (replication 1). In Test 3, PI 598343 had no pod set and in Test 4, PI 533010 (replication 1) and PI 663013 (replication 1), and PI 582468 (replication 3). Due to the substantial amount of rain received during the month of May in 2024, field planting of Test 4 was delayed four weeks. Pod damage or sting marks were observed on all genotypes and replications in each test. Small

rodent damage was observed on the pods of the genotypes in Test 4 during October and were discarded and excluded from the 100-pod subsample.

Percentage of damaged peas

No significant differences in the mean percentage of damaged peas were observed among the genotypes in Test 1 (F=1.42; df=12, 23; $p < .226$). PI 663152 had the lowest mean percentage of damaged peas with 34%, while the susceptible check had the highest mean of 69.6% (Table 2.5). Significant differences were found in Tests 2, 3, and 4. In Test 2, all genotypes except PI 582393 had significantly lower mean percentages of damaged peas than the susceptible check with PI 663152 showing the lowest mean percentage of 19.3% (F=3.18, df=14, 28; $p < .004$). Within groups of genotypes, PI 142779, PI 533009, and PI 663152 had significantly fewer damaged peas than PI 582393, PI 582519, PI 582448, and PI 582468. Additionally, PI 663152 had significantly fewer damaged peas than PI 582512, PI 533010, and PI 598343 (Table 2.6). PI 582393 and the susceptible check were not significantly different. The susceptible check had the highest mean of 67.3% of damaged peas for Test 2 (Figure 2.4).

In Test 3 (F=4.81, df=13, 26; $p < .0003$), PI 664515, PI 663152, 582448, PI 582512, 533010, 582519, PI 142779 and PI 533011 had significantly fewer damaged peas than the susceptible check and PI 533009. Significant differences were also observed within groups of genotypes. PI 664515 and PI 663152, with the lowest percentages of damaged peas (14.6% and 16%), had significantly fewer damaged peas than PI 533011, PI 582349, PI 214354, and PI 582393. PI 664515 also had fewer damaged peas than PI 142779 (Table 2.7). In contrast, the susceptible check and PI 533009 had the highest mean percentages of damaged peas at 67.0% and 67.6% (Figure 2.4).

Test 4 (F=4.79, df=11, 19; p < .001), showed higher percentages of damaged peas compared to the previous tests (Table 2.8). Genotypes PI 582448, PI 582468, PI 582349, and PI 664515 had significantly lower percentages of damaged peas than the three susceptible check plantings, PI 533009, and PI 533010. PI 582519 and PI 214354 had significantly lower mean percentages of damaged peas than PI 533009, with PI 214354 also showing significantly less damage than the susceptible check #2. PI 582448, which had the lowest mean percentage of damaged peas (33.3%), also had a significantly fewer damaged peas than PI 663152, PI 582519, and 214354. The three susceptible checks had mean percentages of 73%, 75.6%, and 70.3% damaged peas, while PI 533009 had the highest mean of 83% (Figure 2.4).

Number of Larvae

Significant differences in the mean number of larvae were observed among the genotypes in Test 1 (F=3.53; df=12, 23; p < .004). PI 582393 had significantly more larvae remaining in pods than most of the genotypes, except for PI 582468 and the susceptible check. Within groups of genotypes, PI 214354, PI 664515, PI 663152, PI 582448, PI 533010 and PI 533011 had significantly fewer larvae than the susceptible check. The genotypes, PI 663152, PI 582448, PI 533010, and PI 533011 also had significantly fewer larvae than PI 582468. No significant differences were found among PI 582393, the susceptible check and PI 582468 (Table 2.5). Both PI 533010 and PI 533011 had zero larvae remaining in pods during shelling. PI 582393 had the highest mean number of larvae at 9.6 followed by the susceptible check with 6.0 (Figure 2.5).

No significant differences in mean number of larvae were detected in Test 2 (F=0.89; df=14, 28; p < .575), Test 3 (F=1.87; df=13, 26; p < .084), or Test 4 (F=1.12; df=11, 19; p < .397). In Test 2, only three genotypes had larvae remaining in pods during shelling. PI 582393 had the highest mean of 0.67, followed by PI 663152 and PI 582468 with 0.33. No larvae were counted

for the susceptible check during this test (Table 2.6). Similar trends were observed in Tests 3 and 4. In Test 3, larvae were recorded in four genotypes, with PI 582393 and the susceptible check showing the highest means at 1.67. Other genotypes with larvae included PI 582468 (1.00) and PI 582519 (0.33) (Table 2.7). In Test 4, most genotypes had low numbers of larvae, except for the susceptible check planting #1. PI 582468 had the highest mean of 3.00, while PI 663152, PI 533010, PI664515, and susceptible check #1, had no larvae remaining in pods during shelling (Table 2.8).

Percentage of damaged pods

No significant differences in the mean percentage of damaged pods were found among the genotypes in Test 1 (F=1.42; df=12, 23; $p < .225$), Test 2 (F=1.38; df=14, 28; $p < .224$), and Test 4 (F=1.94; df=11, 19; $p < .098$). In Test 1, the susceptible check had the highest mean percentage of damaged pods at 100%, while PI 582468 had the lowest mean of 83.3%. For Test 2, PI 582468 again had the lowest mean percentage of damaged pods (82.3%), whereas PI 664515 had the highest mean at 98.3%. In Test 4, PI 582448 had the lowest mean percentage of damaged pods (83.6%), while PI 582468 had the highest mean at 94.5%. Across the three susceptible checks in Test 4, the mean percentages of damaged pods were 90.5%, 90% and 91.3% damaged pods (Figure 2.6).

Significant differences in the mean percentage of damaged pods were observed in Test 3 (F=2.86; df=13, 26; $p < .011$). PI 663152 had significantly fewer damaged pods than all genotypes except PI 582512 and PI 582448. Both PI 582448 and PI 582512 had significantly fewer damaged pods compared to the susceptible check, PI 533010, and PI 533011. Additionally, PI 582448 had a significantly lower percentage of damaged pods than PI 214354 (Table 2.7). Among all the genotypes in Test 3, PI 663152 had the lowest mean percentage of damaged pods at 75.3%, while

the susceptible check had the highest mean of 97% (Figure 2.6). PI 663152, PI 582448, and PI 582512 were the only genotypes that showed significantly lower pod and pea damage than the susceptible check.

Harvest date (Julian date)

No significant differences were observed in the mean harvest dates among the genotypes in Test 1 (F=1.42; df= 12, 23; $p < .225$). Harvest dates ranged from 165.3 JD to 184 JD across all genotypes (Figure 2.7). However, significant differences in mean harvest dates were observed in Tests 2, 3, and 4. In Test 2, the differences were statistically significant (F=2.11; df= 14, 28; $p < .045$), with mean harvest dates ranging from 264 JD to 294 JD (Figure 2.7). Genotypes, PI 533010, PI 533011, and PI 598343 had significantly later mean harvest dates than PI 582349, PI 582512, PI 664515, and the susceptible check. Among these, PI 533011 had the latest mean harvest date, which was significantly later than those of PI 142779, PI 214354, PI 533009, PI 582448, PI 582468, and PI 582519 (Table 2.6).

Test 3 showed a wider range of mean harvest dates compared to Test 1 and 2, spanning from 183.6 JD to 286.5 JD (Figure 2.7). Significant differences were observed, with PI 214354, PI 533010, PI 533011, and PI 582512 having later mean harvest dates than PI 582349, PI 582468, and the susceptible check, PI 582393, PI 663152, PI 582448, and PI 664515, in Test 3 (F=3.48; df= 13, 26; $p <.003$). PI 214354 had a significantly later mean harvest date compared to PI 142779 and PI 582519. Additionally, Test 3 recorded the earliest mean harvest date among the four tests (Table 2.7).

In Test 4, the mean harvest dates ranged from 213 JD to 297 JD, marking the largest range observed across all tests (Figure 2.7). During this test, the susceptible check was planted at staggered dates two and four weeks apart from the other genotypes and the 1st planting of the

susceptible check. Genotypes PI 582468, PI 582448, and PI 582349 showed significantly earlier mean harvest dates than all three of the susceptible check plantings, PI 533010, PI 582519, PI 533009, and PI 214354. Additionally, PI 664515 had a significantly earlier mean harvest date compared to the susceptible check #2, PI 533010, PI 582519 and PI 214354. PI 663152 also showed a significantly earlier mean harvest date compared to PI 582519 (F=5.56 df= 11, 19; $p <$ .0006). Among all the genotypes, PI 533010 had the latest mean harvest date (Table 2.8). Across tests, PI 533010 and Pi 533011 had consistently later harvest dates regardless of planting date.

Yield (clean peas)

Significant differences in the mean yield of clean peas were observed among the genotypes in all four tests. In Test 1, PI 147779 had the highest mean yield of 1005.2 kg/ha, which was significantly greater than all other genotypes (F=2.94; df=12, 23; $p <$ .012). PI 664515 had the lowest mean yield of 385.4 kg/ha, followed by the susceptible check with 390.3 kg/ha (Figure 2.8). No other genotypes showed significant differences in mean yield for Test 1 (Table 2.5).

In Test 2, several genotypes, PI 582519, 142779, 533009, 214254, and PI 582349 produced significantly higher yields than the susceptible check. PI 582519 had the highest yield (2364.9 kg/ha) among all genotypes, significantly higher than all except PI 142779. Differences in mean values were also observed within groups of genotypes (F=4.27; df=14, 28; $p <$ .0005). PI 142779, PI 533009, and PI 214354 had significantly higher yields than PI 663152, PI 664515, PI 582468 and PI 582393. PI 533009 also yielded significantly more than PI 582448 and PI 598343, while PI 142779 yielded significantly more than PI 533010 and PI 533011 (Table 2.6). The susceptible check had the lowest yield of 363.2 kg/ha (Figure 2.8).

In Test 3, PI 582448, PI 664515, PI 582349 had significantly higher mean yields of clean peas than PI 533011, PI 533010, and the susceptible check. PI 582448 and PI 664515 also had

significantly higher mean yields than PI 533009, PI 214354, and PI 582468. PI 582448 with the highest yield (2511.8 kg/ha) produced significantly more than PI 142779 and PI 582393 (F=2.53; df=13, 26; $p < .021$). The susceptible had the lowest yield of clean peas of 308.9 kg/ha (Figure 2.8).

In Test 4, PI 582468 and PI 582519 produced significantly higher mean yields of clean peas than all three of the susceptible checks, as well as PI 663152, PI 533009, and PI 533010. PI 582468 also yielded significantly more clean peas than PI 214354. Significant differences were also observed within groups of genotypes. PI 582448 yielded significantly more than PI 533009, PI 533010 and susceptible checks #2 and #3. PI 582349 yielded significantly more than susceptible #3 (F=3.04; df=11, 19; $p < .016$). PI 582468 had the highest yield of clean peas of 1062.9 kg/ha, while 533010 had the lowest yield of clean peas of 75.4 kg/ha. The three susceptible checks produced consistently low yields of clean peas at 126.7 kg/ha, 122.6 kg/ha, and 107.7 kg/ha, with no significant differences among them (Figure 2.8).

Discussion

The objective of the open choice experiments were to evaluate insect damage on the different genotypes in a natural field environment which provides insight into cowpea curculio host preference and potential mechanisms of resistance of the genotypes across multiple tests. The genotypes selected for these experiments exhibit a broad range of variability in growth patterns, morphology, maturity, and seed characteristics. In addition to variability among the different genotypes, the environmental factors and cowpea curculio populations were also variables changing between tests which can affect each of the genotypes differently. Factors such as the variability in maturity time, pod development, and peak population levels of cowpea curculios must be considered when assessing damage. A genotype might show less damage if its pod development coincides with a period of low cowpea curculio activity, compared to a genotype with

a pod set during peak cowpea curculio activity. In Tifton, trapped adult counts for two cowpea curculio generations have been reported to increase approximately five-fold from the May-June peak to the August-September peak (Riley et al., 2015). Considering the published data, there should be two cycles of peak cowpea curculio activity during the first three open choice field tests and three cycles during open choice field Test 4.

The variability in weather conditions may have contributed to the greater damage observed in Test 4 compared to the three previous tests. The severity of damage in this test could be a combination of the delayed planting and the availability of pods for starving, overwintered cowpea curculios. Since cowpea is the main reproductive host for cowpea curculios, the first emerging generation had limited food options and essentially no reproductive crop. Cowpea curculios will feed on various parts of the cowpea plant, such as leaves and flower buds, until pods become available (Riley & Sparks, 2014). The number of pods available for harvest in many plots was limited, possibly due to the destruction of flowers by the first generation of cowpea curculios. As an indeterminate crop, the cowpea genotypes eventually began to flower again, and this could have contributed to the increased number of harvest days. The data for PI 598343 are inconclusive due to the lack of pod set in two of the three experiments. When the genotype did set pods, the planting was in July which may reflect the genotype's photoperiod sensitivity.

The results of this study indicated varying levels of resistance reactions to the cowpea curculio among the genotypes, primarily through antixenosis (or reduced preference) and/or antibiosis (or reduced reproduction) resistance mechanisms. Data of the percentages of damaged peas support the resistance potential of certain genotypes. In Tests 2, 3, and 4, significant differences were observed with PI 663152, PI 664515, and PI 582448 consistently exhibiting the lowest percentages of damaged peas. The parental lineages of PI 663152 and PI 664515 have

previously been associated with high pod wall factors and resistance to cowpea curculio through antixenosis (Cuthbert, 1976; Fery et al., 1993). Results confirm this association is most likely present in these genotypes. Additionally, PI 664515 has a small seed size that may contribute to a feeding and oviposition deterrence, but further experiments are needed to confirm this. The variability and severity of damage shown by PI 633152 and PI 664515, especially PI 663152 in Test 4, possibly suggests that the two genotypes may have a threshold of insect pressure each are able to withstand, or the cowpea curculios are becoming resistant to the factors of resistance expressed by these two genotypes. Other genotypes that showed lower percentages of damaged peas, possibly expressing resistance, included PI 142779, PI 214354, PI 533010, PI 533011, PI 582512, and PI 582349. Although percentages in damage varied across tests, these genotypes consistently had less than 40% of damaged peas and showed significantly lower percentages of damaged peas than the susceptible check (67-75.6%) in either one or two of the tests.

Only Test 3 revealed significant differences in percentages of damaged pods, with PI 663152, PI 582448, and PI 582512 showing significantly fewer damaged pods compared to the susceptible check. These results suggest antixenosis resistance mechanism is present in these genotypes. Additionally, these genotypes were also the only ones to show significantly fewer damaged pods and peas than the susceptible check for Test 3. Although, results only provide evidence in a single test for lower pod damaged, the data for lower percentages of damaged peas further supports the presence of a resistance mechanism in these genotypes. Lower percentages of pod and pea damage indicate that these genotypes are less preferred for feeding and oviposition, possessing physical traits, such as pod wall thickness that may function as barriers, reducing seed damage (Chambliss & Hunter, 1997). These results highlight the genotype's potential of providing sources of resistance and/or additional selections for future research on cowpea curculio.

The larval counts aligned with percentages of pod and pea damage observations, where resistant genotypes had fewer or no larvae remaining in the pods. While significant differences were primarily observed in Test 1, larval counts across tests suggests some trends among certain genotypes, specifically with presence of larvae remaining in pods. This indicates a lack or low level of antibiosis resistance present. PI 533010 and PI 533011 were the only genotypes with no larvae counted suggesting potential antibiosis resistance, as well as a possible low level of antixenosis resistance indicated by the low percentages of damaged peas in either one or two of the tests. However, due, to the small numbers of larvae counted in three out of four tests, there is not enough data and/or evidence to confidently suggests that either one of the genotypes have antibiosis resistance from the results of a single test.

The measurement for clean pea yield is more complex due to the differences in pea size and weight, and the number of pods harvested for each of the genotypes, which can affect over all yield results, but this does provide insights into each genotype's performance under cowpea curculio pressure as well as other variability in environmental conditions that occurred during each test. The agronomic benefits of resistance expressed by the genotypes were reflected by clean pea yield data. Genotypes with lower percentages of damaged pod and peas, such as PI 582448, PI 142779, and PI 582519, produced significantly higher yields than the susceptible check. Consistently producing higher yields under the varying levels insect pressure across each test highlight the potential for sustaining productivity in areas heavily impacted by the cowpea curculio. This also suggests that these genotypes may have a level of tolerance to withstand or recover from insect attack and/or pressure of cowpea curculio populations.

Overall, the resistance reactions observed in the results of the open choice experiments provided insight into the cowpea curculio's preference of genotype, as well as the level of

resistance each of the genotypes possesses towards the cowpea curculio. Resistance factors in cowpeas that have been documented include pod wall thickness or toughness serving as barriers to feeding and/or oviposition preventing penetration by adult female insects (Chambliss & Rymal, 1980), biochemical compounds acting as feeding deterrents such as increased levels of tannins or lower quantities of water soluble feeding stimulants (Chalfant & Gaines, 1973; Rymal & Chambliss, 1976), and different levels or types of volatile compounds acting as non-attractants (Chambliss & Rymal, 1982). The resistance exhibited in variable environmental conditions and fluctuations in cowpea curculio activity suggest that the genotypes, PI 663152, PI 582448, and PI 664515, have potential use in breeding programs focused on cowpea curculio resistance. Further research could investigate the genetic basis of resistance and validate these findings.

Tables and Figures

Table 2.1. Genotypes selected for evaluation of cowpea curculio resistance.

Treatment No.	Genotype ID (PI No.)	Other Names	Origin
1	PI 142779	CUARENTANA	Chiapas, Mexico
2	PI 214354	TVu 1570	Punjab, India
3	PI 533009	TARS-36	Puerto Rico, United States
4	PI 533010	TARS-62	Puerto Rico, United States
5	PI 533011	TARS-48	Puerto Rico, United States
6	PI 582349	CPI 30783	Myanmar
7	PI 582393	IT82E-18	Nigeria
8	PI 582448	UCR 306; IT82D-713	Nigeria
9	PI 582468	UCR 347; Mn 150	Minnesota, United States
10	PI 582512	IT84S-2049	Nigeria
11	PI 582519	IT84S-2246-4	Nigeria
13	PI 663152	CR 22-2-21	South Carolina, United States
14	PI 664515	Bettergreen	South Carolina, United States
15	PI 663013 (Susc.)	California Blackeye 5	California, United States

Table 2.2. Genotypes selected for evaluation of cowpea curculio resistance in Test 4.

Treatment No.	Genotype ID (PI No.)	Other Names	Origin
2	PI 214354	TVu 1570	Punjab, India
3	PI 533009	TARS-36	Puerto Rico, United States
4	PI 533010	TARS-62	Puerto Rico, United States
6	PI 582349	CPI 30783	Myanmar
8	PI 582448	UCR 306; IT82D-713	Nigeria
9	PI 582468	UCR 347; Mn 150	Minnesota, United States
11	PI 582519	IT84S-2246-4	Nigeria
13	PI 663152	CR 22-2-21	South Carolina, United States
14	PI 664515	Bettergreen	South Carolina, United States
15	PI 663013 (Susc.)	California Blackeye 5	California, United States

Table 2.3. ANOVA output results for parameters measured in evaluations of cowpea curculio damage on genotypes from Test 1 and Test 2 in 2022.

Parameters	**Test 1**			**Test 2**		
	DF(n), df(d)	*F*	Pr>F	DF(n), df(d)	*F*	Pr>F
Number of pods/4.572 m	12, 23	2.52	.027	14, 28	3.53	.002
Weight of pods (g)	12, 23	4.74	.0007	14, 28	7.96	<.0001
Percentage of damaged pods	12, 23	1.42	.225	14, 28	1.38	.224
Weight of clean pods (g)	12, 23	1.22	.326	14, 28	2.91	.007
Weight of damage pods (g)	12, 23	4.91	.0005	14, 28	8.00	<.0001
Weight of peas (g)	12, 23	5.05	.0004	14, 28	5.72	<.0001
Weight of clean peas (g)	12, 23	0.67	.763	14, 28	3.21	.004
Weight of damaged peas (g)	12, 23	4.61	.0008	14, 28	5.86	<.0001
Percentage of damaged peas	12, 23	1.42	.226	14, 28	3.18	.004
Number of larvae counted	12, 23	3.53	.004	14, 28	0.89	0.575
Clean pea yield (kg/ha)	12, 23	2.94	.012	14, 28	4.27	.0005
Harvest date (Julian date)	12, 23	1.42	.225	14, 28	2.11	.045

Table 2.4. ANOVA output results for parameters measured in evaluations of cowpea curculio damage on genotypes from Test 3 in 2023 and Test 4 in 2024.

Parameters	**Test 3**			**Test 4**		
	DF(n), df(d)	*F*	Pr>F	DF(n), df(d)	*F*	Pr>F
Number of pods/4.572 m	13, 26	3.73	.002	11, 19	3.01	.017
Weight of pods (g)	13, 26	9.06	<.0001	11, 19	5.71	.0005
Percentage of damaged pods	13, 26	2.86	.011	11, 19	1.94	.098
Weight of clean pods (g)	13, 26	1.92	.076	11, 19	7.72	<.0001
Weight of damage pods (g)	13, 26	10.25	<.0001	11, 19	4.86	.001
Weight of peas (g)	13, 26	1.54	.167	11, 19	2.23	.060
Weight of clean peas (g)	13, 26	2.59	.018	11, 19	7.68	<.0001
Weight of damaged peas (g)	13, 26	2.03	.059	11, 19	2.08	.077
Percentage of damaged peas	13, 26	4.81	.0003	11, 19	4.79	.001
Number of larvae counted	13, 26	1.87	.084	11, 19	1.12	.397
Clean pea yield (kg/ha)	13, 26	2.53	.021	11, 19	3.04	.016
Harvest date (Julian date)	13, 26	3.48	.003	11, 19	5.56	.0006

Table 2.5. Means of parameters measured in evaluations on the different genotypes from Test 1 in April 2022.

Genotypes	Parameters									
	# pods/4.572 m	% damaged pods	Clean pod wt (g)	Damaged pod wt (g)	Clean pea wt (g)	Damaged pea wt (g)	% damaged peas	# of larvae	Clean pea yield (kg/ha)	Harvest date (JD)
PI 142779	511.6a	90.33	38.67	401.6bcd	8.33	164.6b	52.00	3.00bcd	1005.29a	174.3
PI 214354	358.6bc	89.00	41.33	335.6bcde	11.33	109.3bcde	49.33	1.33cd	550.22b	174.0
PI 533009	234.0c	87.67	57.00	587.6a	12.33	242.0a	47.00	1.67bcd	669.82b	168.6
PI 533010	325.5bc	90.50	51.50	299.5cde	12.00	142.5bcd	50.50	0.00d	639.33b	184.0
PI 533011	271.0bc	95.33	28.00	366.0bcd	6.00	153.0bc	47.33	0.00d	545.51b	174.6
PI 582349	361.6bc	84.00	67.00	316.0cde	10.33	91.6cde	56.33	2.00bcd	427.15b	172.3
PI 582393	402.0ab	86.33	60.00	386.0bcd	8.00	131.3bcd	56.67	9.67a	591.13b	165.3
PI 582448	397.0ab	90.00	24.67	280.3de	6.67	83.67de	43.33	0.33d	497.54b	170.6
PI 582468	286.0bc	83.33	35.33	229.3e	11.00	108.6bcde	57.33	5.33abc	397.22b	165.3
PI 582512	294.0bc	92.00	28.00	420.3bc	5.33	156.0bc	52.33	1.67bcd	542.83b	176.6
PI 582519	-	-	-	-	-	-	-	-	-	-
PI 598343	-	-	-	-	-	-	-	-	-	-
PI 663152	298.0bc	90.67	19.33	227.0e	21.00	53.3e	34.00	0.67d	394.09b	179.0
PI 664515	310.6bc	88.67	37.67	402.3bcd	6.33	100.3bcde	55.67	1.00cd	385.46b	176.0
PI 663013 (Susc.)	336.0bc	100.00	0.00	451.3b	0.00	156.6bc	69.67	6.00ab	390.39b	166.0
LSD	134.82	11.05	49.48	130.9	17.88	65.97	21.11	4.50	295.06	12.88

*Means within columns followed by the same letter are not significantly different (P<0.05).

Table 2.6. Means of parameters measured in evaluations on the different genotypes from Test 2 in July 2022.

Genotypes	Parameters									
	# pods/4.572 m	% damaged pods	Clean pod wt (g)	Damaged pod wt (g)	Clean pea wt (g)	Damaged pea wt (g)	% damaged peas	# of larvae	Clean pea yield (kg/ha)	Harvest date (JD)
PI 142779	338.7cde	87.67	97.3abc	715.7b	24.6a	252.6bc	27.6cd	0.00	1768.8ab	272.0bcd
PI 214354	412.7bcde	92.33	32.3ef	428.3de	6.3b	195.3bcd	35.6bcd	0.00	1356.7bcd	274.3bcd
PI 533009	192.7e	89.67	103.3ab	1003.3a	26.0a	397.3a	26.6cd	0.00	1447.0bc	276.6bcd
PI 533010	214.0e	84.00	113.3a	660.0bc	24.0a	268.0b	41.3bc	0.00	821.6cdef	283.3abc
PI 533011	169.0e	95.33	68.3abcde	710.7b	13.0ab	277.6b	32.0bcd	0.00	821.6cdef	294.0a
PI 582349	567.3abcd	89.33	39.6def	409.7de	8.0b	144.6de	36.0bcd	0.00	1211.6bcde	264.0d
PI 582393	332.0cde	87.00	45.3cdef	316.3de	7.6b	109.3de	50.6ab	0.67	431.6ef	280.0abcd
PI 582448	592.0abc	88.67	28.6ef	272.0de	7.0b	83.3e	47.6b	0.00	627.6def	274.0bcd
PI 582468	208.3e	82.33	61.6abcde	276.3de	15.0ab	152.0cde	47.0b	0.33	465.0ef	274.0bcd
PI 582512	663.3ab	85.33	49.3bcdef	333.0de	8.0b	100.0de	42.0bc	0.00	1019.0bcdef	265.3d
PI 582519	825.3a	85.00	73.6abcde	447.0cde	25.6a	190.0bcd	48.3b	0.00	2364.9a	270.0bcd
PI 598343	281.7de	92.33	92.3abcd	256.3e	27.0a	116.6de	39.6bc	0.00	620.0def	285.6ab
PI 663152	287.7de	92.33	24.3ef	274.3de	6.0b	92.6de	19.3d	0.33	517.3ef	279.6abcd
PI 664515	285.0de	98.33	5.0f	480.3cd	0.6b	124.0de	37.6bcd	0.00	500.1ef	268.0cd
PI 663013 (Susc.)	291.7cde	94.00	23.6ef	420.3de	6.6b	146.6de	67.3a	0.00	363.2f	264.0d
LSD	301.55	11.12	56.38	220.6	14.84	105.41	18.94	0.60	807.65	16.97

*Means within columns followed by the same letter are not significantly different ($P<0.05$).

Table 2.7. Means of parameters measured in evaluations on the different genotypes from Test 3 in 2023.

Genotypes	Parameters									
	# pods/6.096 m	% damaged pods	Clean pod wt (g)	Damaged pod wt (g)	Clean pea wt (g)	Damaged pea wt (g)	% damaged peas	# of larvae	Clean pea yield (kg/ha)	Harvest date (JD)
PI 142779	426.7bcdef	86.6bcd	68.6	478.0defg	20.8bcd	193.7	36.3cd	0.00	1108.7bcd	220.3bc
PI 214354	304.7cdef	91.0abc	44.3	451.6efg	12.0cd	168.3	47.3abc	0.00	589.9cd	286.5a
PI 533009	327.7cdef	89.3abcd	64.1	680.2ab	11.3cd	258.2	67.6a	0.00	513.3cd	235.3abc
PI 533010	209.3ef	94.0ab	32.0	649.4abc	6.4d	150.1	32.6cde	0.00	395.3d	271.6ab
PI 533011	173.3f	91.6ab	55.5	738.3a	7.4d	208.4	39.6bc	0.00	402.6d	271.6ab
PI 582349	567.0bc	85.0bcd	91.6	519.3def	36.5ab	210.6	46.3abc	0.00	1783.8abc	183.6c
PI 582393	566.7bc	88.6abcd	60.4	439.4efg	20.7bcd	184.7	48.0abc	1.67	1087.6bcd	208.8c
PI 582448	874.0a	81.0de	86.3	389.4g	44.5a	152.7	29.0cde	0.00	2511.8a	208.8c
PI 582468	340.7bcdef	88.0abcd	54.5	416.6fg	26.8abcdb	182.6	58.3ab	1.00	629.0cd	183.6c
PI 582512	474.3bcde	82.0cde	107.1	490.0defg	31.1abc	182.2	32.6cde	0.00	1307.5abcd	271.6ab
PI 582519	460.0bcde	90.0abcd	64.6	574.4bcd	21.8abcd	253.8	35.0cde	0.33	1552.1abcd	220.1bc
PI 598343	-	-	-	-	-	-	-	-	-	-
PI 663152	617.3ab	75.3e	88.5	260.9h	37.7ab	124.8	16.0de	0.00	1583.4abcd	208.8c
PI 664515	534.3bcd	85.3bcd	97.9	527.2cde	31.4abc	216.3	14.6e	0.00	2125.5ab	211.5c
PI 663013(Susc.)	252.0ef	97.0a	14.4	550.7cde	3.6d	195.3	67.0a	1.67	308.9d	207.5c
LSD	284.23	9.60	54.69	113.09	23.35	75.92	21.63	1.33	1283.9	52.97

*Means within columns followed by the same letter are not significantly different (P<0.05).

Table 2.8. Means of parameters measured in evaluations on the different genotypes from Test 4 in 2024.

Genotypes	Parameters									
	# pods/4.572 m	% damaged pods	Clean pod wt (g)	Damaged pod wt (g)	Clean pea wt (g)	Damaged pea wt (g)	% damaged peas	# of larvae	Clean pea yield (kg/ha)	Harvest date (JD)
PI 214354	527.6a	90.0	29.6b	273.0de	5.5c	71.5	53.6cde	0.33	458.7bcde	272.6abc
PI 533009	160.6b	90.0	53.3a	518.8ab	13.3b	166.7	83.0a	0.33	129.3cde	284.3ab
PI 533010	112.5b	87.0	17.1b	348.4cde	5.3c	130.8	75.5ab	0.00	75.4e	297.0a
PI 582349	331.3ab	87.3	49.5a	480.2abc	9.0bc	119.9	47.6de	0.33	627.6abcd	226.9de
PI 582448	238.3b	83.6	64.0a	368.2bcde	20.9a	144.1	33.3e	0.67	668.0abc	221.3de
PI 582468	306.0ab	94.5	29.7b	511.8ab	12.2b	233.9	46.0de	3.00	1062.9a	213.0e
PI 582519	520.3a	85.6	53.5a	381.1bcd	13.3b	124.9	58.6bcd	0.33	842.8ab	287.6a
PI 663152	247.0b	90.6	19.4b	214.6e	3.7c	68.6	64.6abcd	0.00	167.4cde	251.0bcd
PI 664515	209.3b	87.6	57.2a	553.0a	12.0b	175.4	46.3de	0.00	537.2abcde	244.8cde
PI 663013 (Susc. #1)	108.5b	90.5	14.1b	394.4bcd	3.9c	168.9	73.0abc	0.00	126.7cde	283.0ab
PI 663013 (Susc. #2)	199.3b	90.0	20.3b	294.1de	3.4c	77.6	75.6ab	1.33	122.5cde	276.8abc
PI 663013 (Susc. #3)	128.7b	91.3	21.4b	278.4de	4.2c	106.4	70.3abc	1.33	107.7de	284.6ab
LSD	250.3	6.15	19.54	153.9	6.04	97.86	21.16	2.18	551.14	36.31

*Means within columns followed by the same letter are not significantly different (P<0.05).

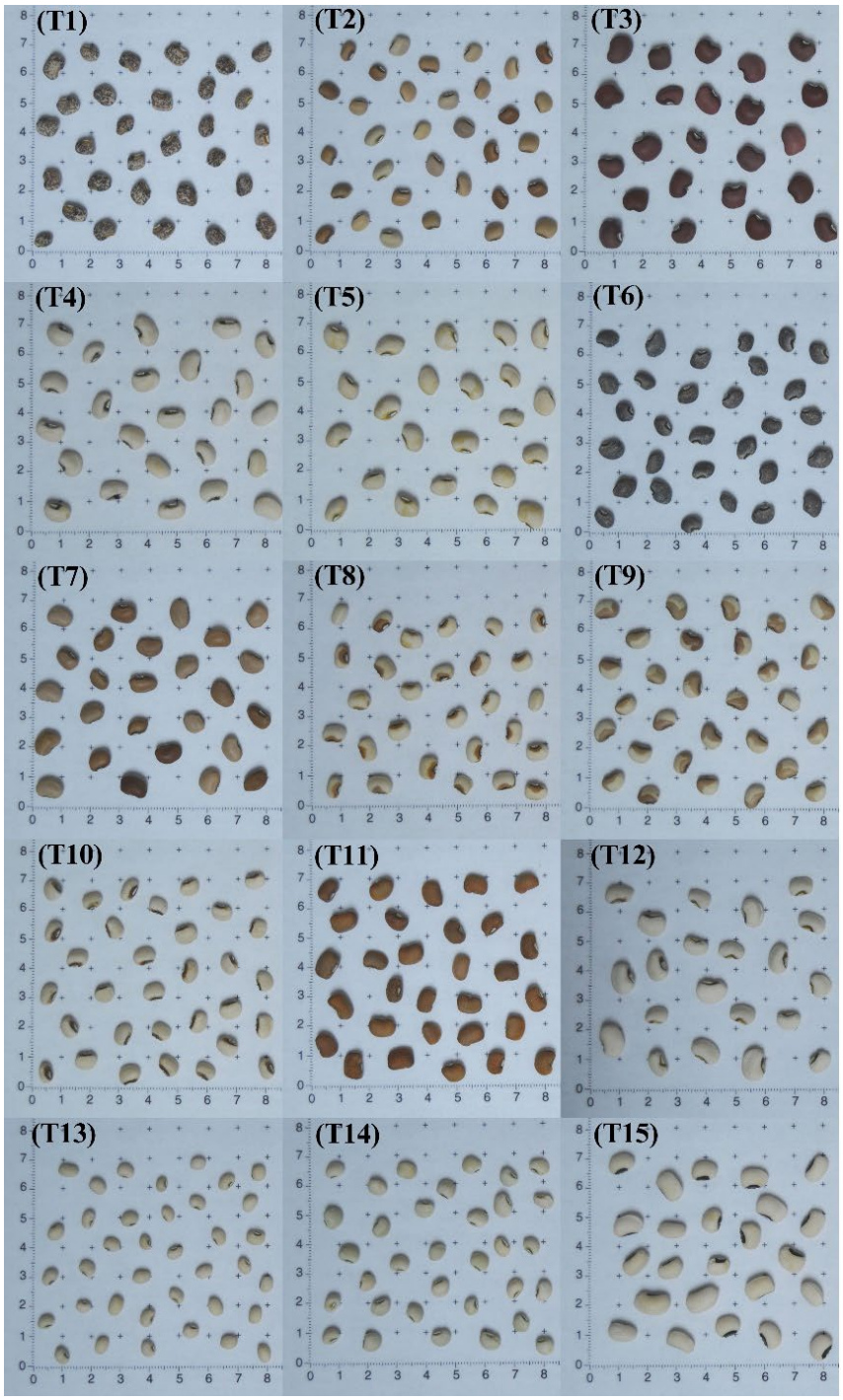

Figure 2.1. Genotypes labeled by treatment number; T1: PI 142779, T2: PI 214354, T3: PI 533009, T4: PI 533010, T5: PI 533011, T6: PI 582349, T7: PI 582393, T8: PI 582448, T9: PI 582468, T10: PI 582512, T11: PI 582519, T12: PI 598343, T13: PI 663152, T14: PI 664515, T15: PI 663013 (susceptible).

Figure 2.2. Cowpea curculio “Sting” damage on fresh green pod (cm); (A): PI 582393, (B): PI 214354, (C) PI 582519, (D): PI 663013 (susceptible).

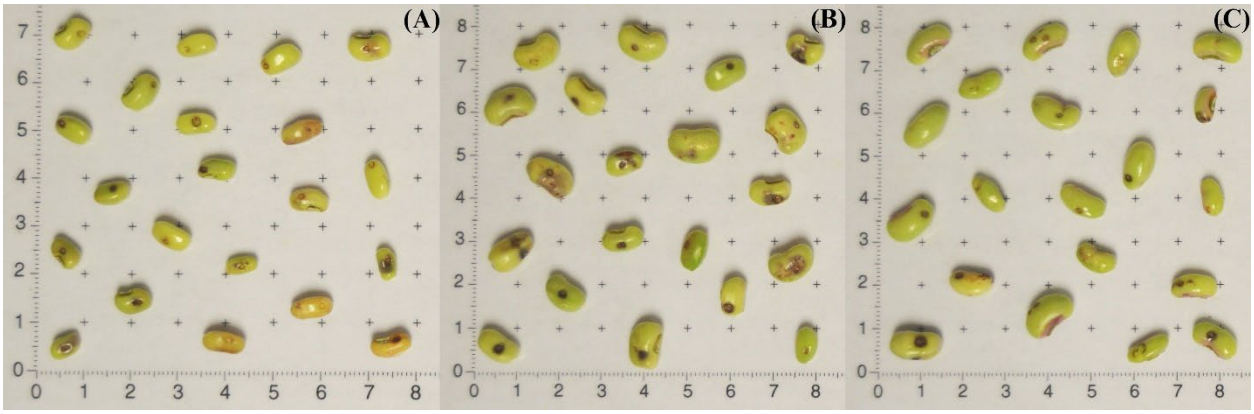

Figure 2.3. Cowpea curculio “Sting” damage on fresh green seed (cm); (A): PI 214354, (B): PI 582519, (C): PI 663013 (susceptible).

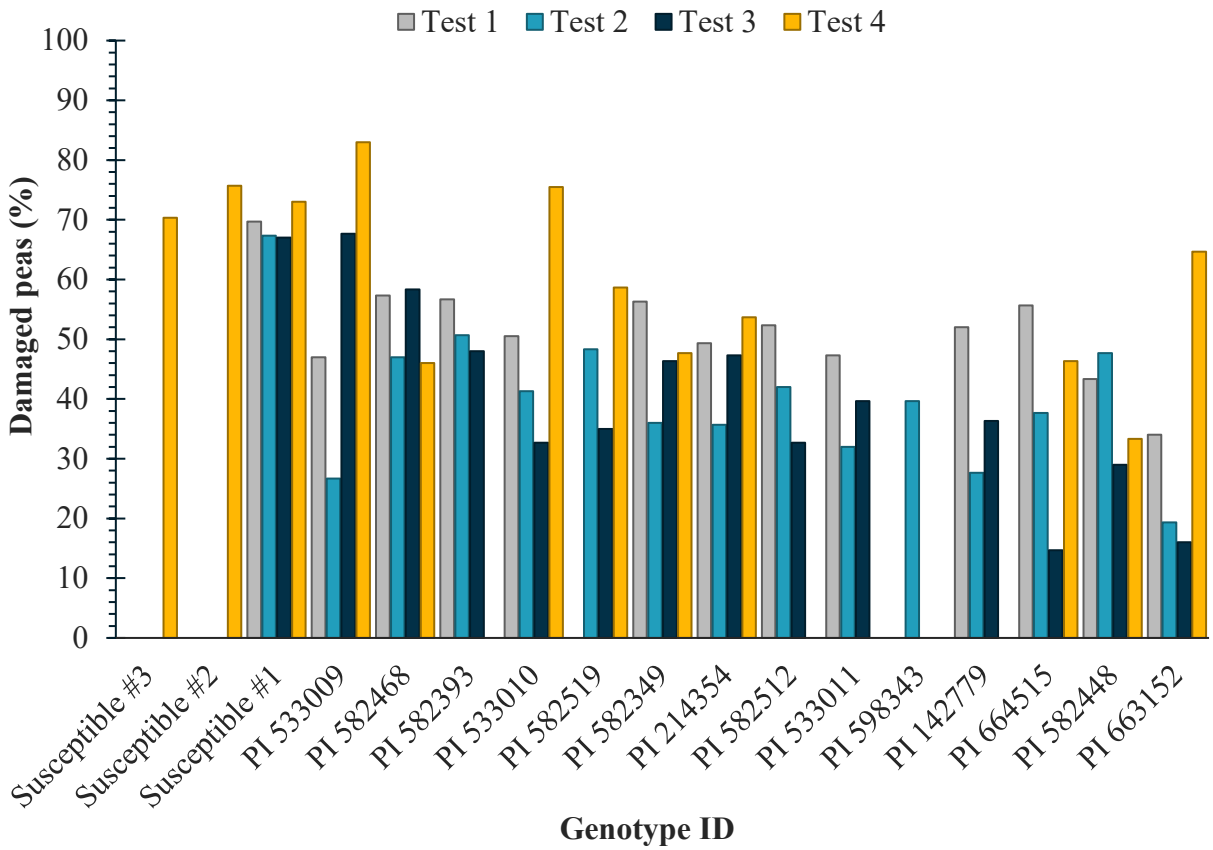

Figure 2.4. Bar plot showing genotype mean percentage of damaged peas in open choice experiments from Test 1, Test 2, Test 3, and Test 4.

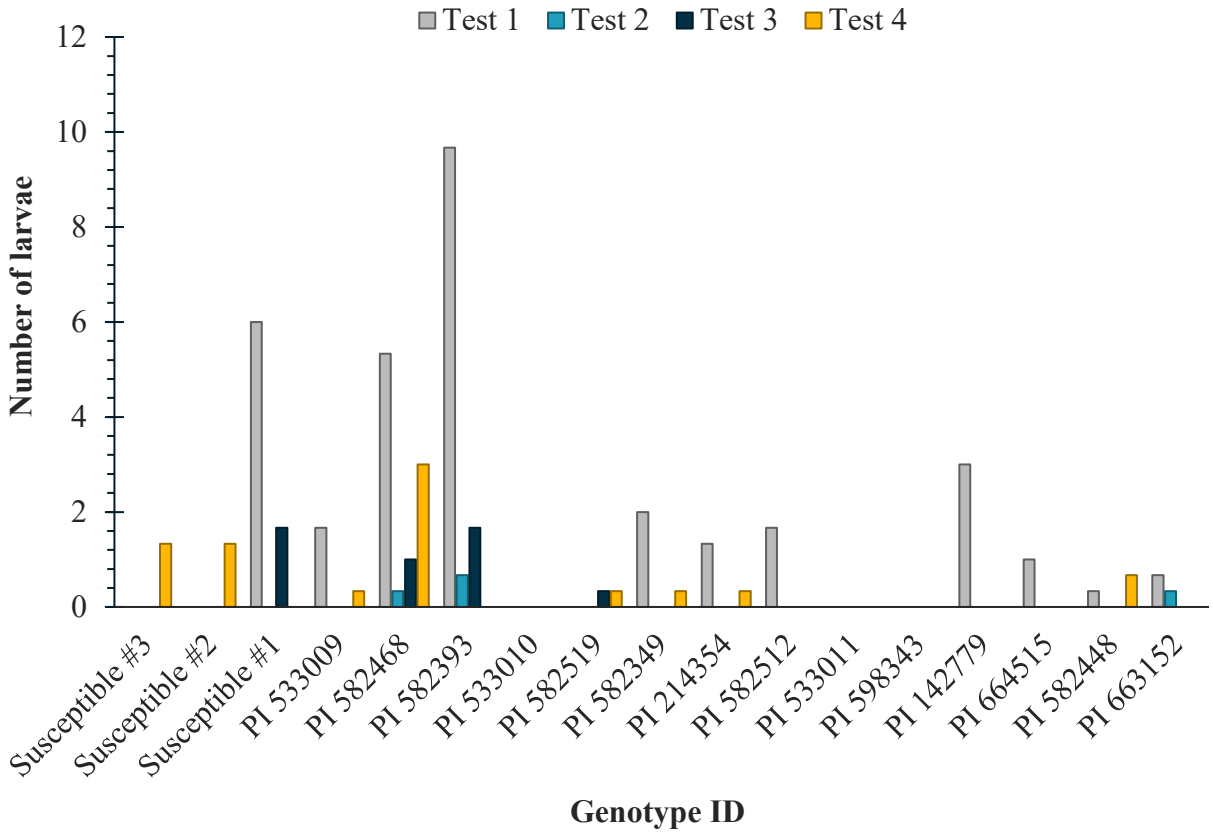

Figure 2.5. Bar plot showing genotype mean number of larvae remaining in pods in open choice experiments from Test 1, Test 2, Test 3, and Test 4.

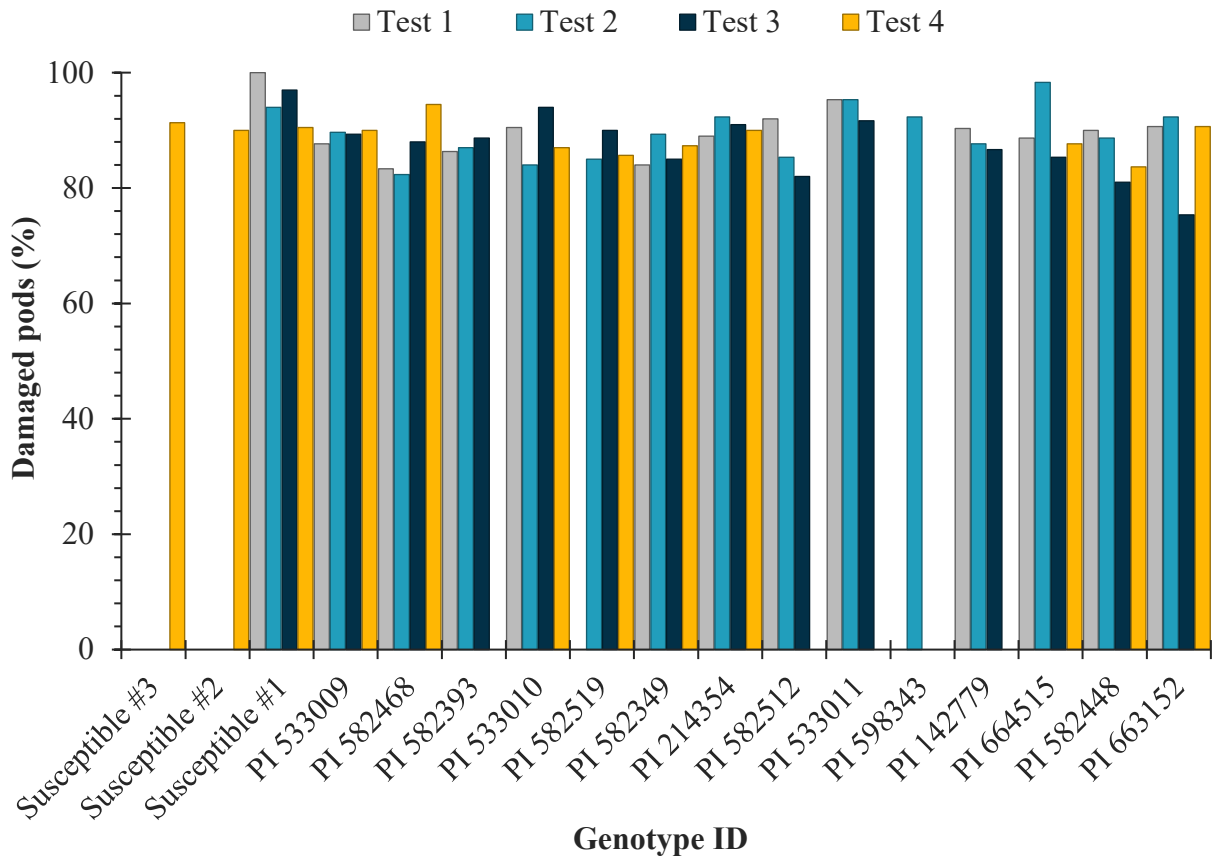

Figure 2.6. Bar plot showing genotype mean percentage of damaged pods in open choice experiments from Test 1, Test 2, Test 3, and Test 4.

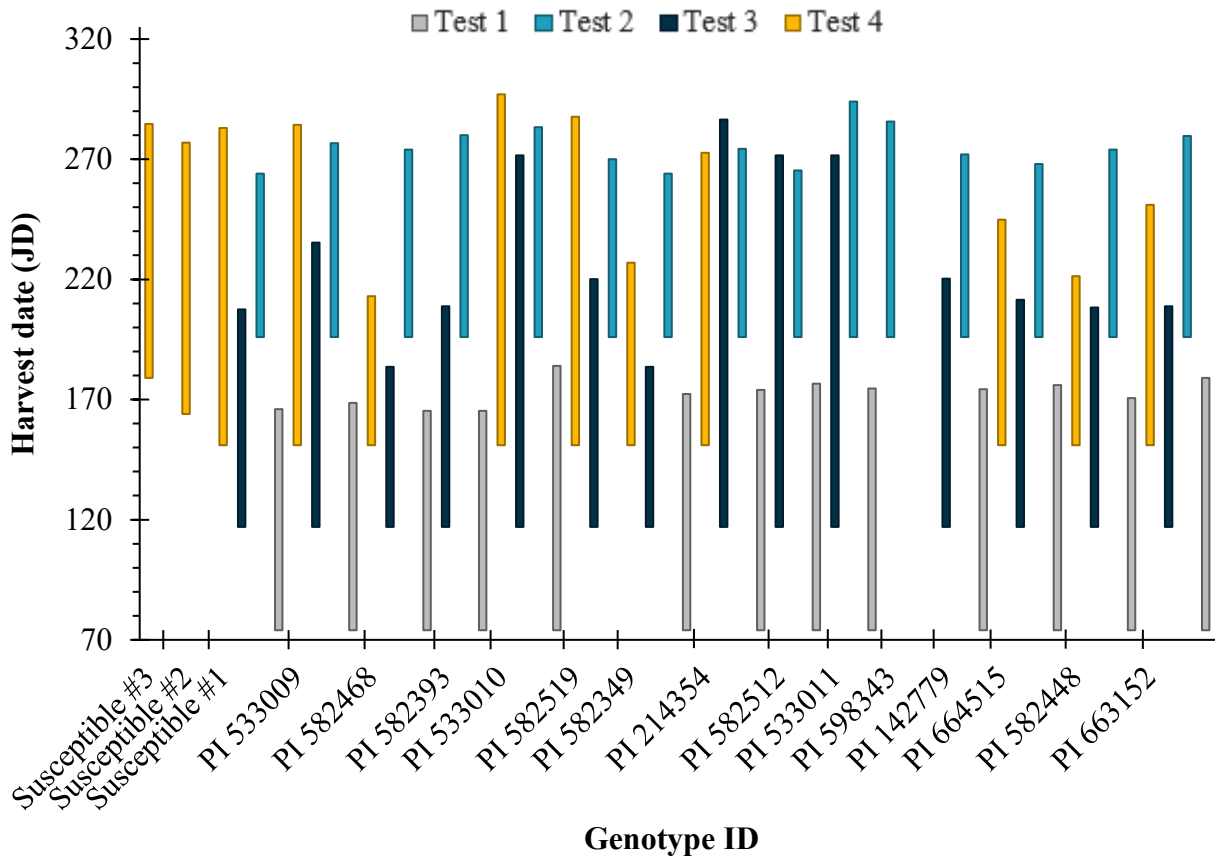

Figure 2.7. Bar plot showing genotype mean harvest date (Julian date) in open choice experiments from Test 1, Test 2, Test 3, and Test 4.

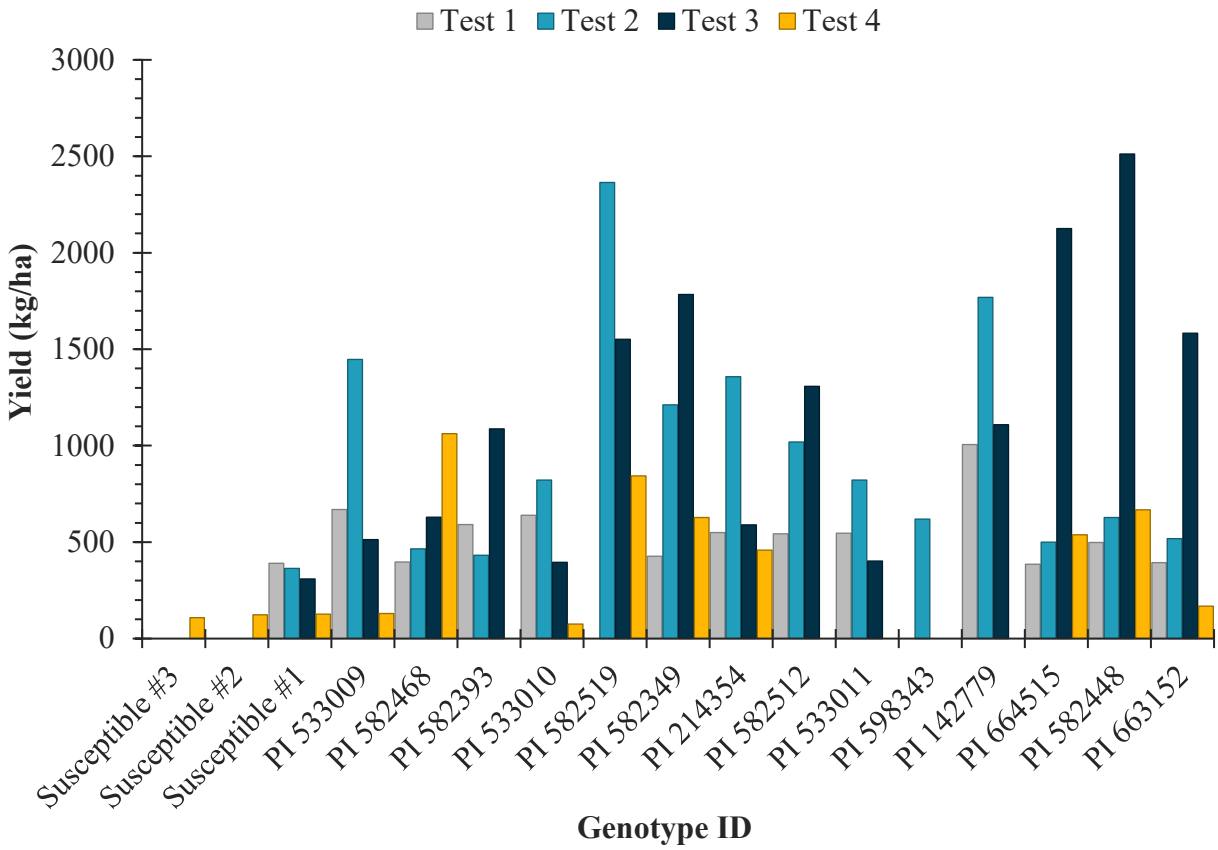

Figure 2.8. Bar plot showing genotype mean yield of clean peas (kg/ha) in open choice experiments from Test 1, Test 2, Test 3, and Test 4.

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CHAPTER 3

EVALUATION OF COWPEA GENOTYPES FOR RESISTANCE TO THE COWPEA CURCULIO IN A NO CHOICE EXPERIMENT[1]

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Abstract

Cowpea, *Vigna unguiculata* L. Walp., is an important crop in the southern United States, but its production is severely impacted by the cowpea curculio, *Chalcodermus aeneus* Boheman, a major pest that causes significant pod and seed damage. Developing resistant cowpea cultivars is essential for minimizing yield losses and reducing reliance on chemical control. Based on the results from Tests 1 and 2 of the open choice experiments (Chapter 2), five genotypes, PI 533009, PI 533010, PI 582349, PI 582468, and PI 663152, along with the susceptible check were selected for a no choice experiment to further evaluate genotypic differences in resistance. In this study, adult cowpea curculios were restricted to individual genotypes in a controlled environment to evaluate insect preference when presented with a single option for feeding and oviposition. Pod and pea damaged were assessed to determine level of resistance if present. While there were no significant differences in the number of sting marks on the pods and the percentage of adult female cowpea curculios, there were significant differences observed among the genotypes for mean pod and pea weight and the percentage of damaged peas. The significant results observed among the genotypes, with some exhibiting less damage, suggests potential resistance mechanisms, such as antixenosis and/or antibiosis. These findings contribute to efforts to identify resistant germplasm for cowpea breeding programs and the search for control of the cowpea curculio.

Introduction

One of the primary goals of this project is to identify the presence of resistance in cowpea germplasm to the cowpea curculio, *Chalcodermus aeneus* Boheman. The cowpea curculio is a small weevil that primarily affects leguminous crops, such as cowpea and other alternate host plants (Sudbrink et al., 1998). It is a major insect pest, especially in the southern United States, where cowpeas are a staple crop. The cowpea curculio is among the most destructive pests,

targeting developing pods and seeds, leading to substantial yield losses (Riley & Sparks, 2019). The cowpea curculio's impact has escalated in recent years, severely limiting production and threatening the sustainability of this crop in the southeastern region. Cowpea was a major agronomic crop in the United States in 1937 when production area peaked at 2.4 million ha, placing the United States in the top 10 global producers until 1967 (Osipitan et al., 2021). In 2023, production area has drastically declined to 7280 ha (FAOSTAT, 2023). There could be a combination of reasons for this decline, but the inability to consistently manage or control the cowpea curculio in the main production areas of the southeast was one of the main reasons for the decline in those areas (Riley & Sparks, 2019).

Current management practices for cowpea curculio, such as chemical insecticides are often insufficient and costly. Host plant resistance offers a promising, sustainable alternative for controlling cowpea curculio infestations (Togola et al., 2017). Developing resistant genotypes not only reduces reliance on chemical controls but also enhances the long-term viability of cowpea production systems. Identifying and incorporating resistance traits into breeding programs is essential to alleviate this pest's impact on cowpea production in the region and prevent further impacts of infestation to other growing regions in the country.

No choice bioassays are conducted to evaluate insect resistance in controlled environments. This method confines insects to a single host genotype, eliminating any alternate options to assess resistance traits for each genotype independently (Van Driesche and Murray, 2004). By minimizing external variables, no choice experiments provide accurate data on traits affecting feeding, oviposition, larval development, and survival. These experiments are effective for assessing both antixenosis and antibiosis resistance mechanisms. No choice tests have been widely used alongside open choice tests to identify and confirm insect resistance in plants.

The goal of this study is to evaluate the resistance of the selected genotypes to the cowpea curculio using no choice bioassays. Genotypes were selected from open choice field evaluations to further evaluate genotypic differences in pod and pea damage. The bioassay experiments provide additional information on genotype resistance by evaluating cowpea curculio damage in a controlled environment with a single host choice compared to a variable field environment with open choice. By identifying genotypes with resistance, this research will contribute to the development of cultivars resistant to the cowpea curculio. Results will provide insights into host plant resistance mechanisms and support breeding efforts to improve cowpea production in the southeastern United States.

Materials and Methods

Plant material

Five genotypes and a susceptible check were selected from the fourteen genotypes evaluated in the open choice field experiments (Table 3.1). These were selected to further evaluate genotypic differences among pod and pea damage for resistance to the cowpea curculio. All materials were obtained from the USDA-ARS cowpea germplasm collection in Griffin, GA.

Experimental design

The no choice experiment was conducted at the University of Georgia, Griffin campus in the USDA screenhouse. Genotypes were planted on June 30th, 2023, and July 7th, 2023, to coordinate pod maturity and cowpea curculio peak population levels in the field. Each of the genotypes were planted in three-gallon plastic potting containers. Potting containers were arranged in a randomized complete block design with six replications per entry and one plant per pot. Cowpea curculio adults were removed from a food source for a 24-hour period prior to bioassays. Insect rearing bags (30cm by 20cm) were secured around two intact pods with four adults for

feeding and ovipositioning (Figure 3.1). Each replication was sampled up to eight times. After the 72 hours, the cowpea curculios were removed and placed in the freezer to later determine sex of the adults. Cowpea curculios were placed in 70% ETOH and dissected under a microscope where female reproductive structures could be examined. The weight of the sampled pods (g), number of stings on pods, weight of peas (g), and cowpea curculio sex were recorded, and the totals from samples were averaged for each replication per entry. All data were analyzed using Proc GLM in SAS and mean separation determined by LSD ($P<0.05$).

Source and handling of cowpea curculio adults

Adult cowpea curculios were collected by Modified Tedders traps (Figure 3.2) on the Lang-Rigdon Farm Hook Field at the University of Georgia, Tifton Campus in Tifton, Georgia (Latitude 31.514749, Longitude -83.547925). Traps were used to collect cowpea curculios from August 2023 to September 2023. Each trap was placed in the field adjacent to a known susceptible cowpea cultivar during pod development. After cowpea curculios were trapped and collected, insects were maintained on a diet of fresh green pods until removed for the no choice experiment. The fresh green pods used as a food source were the same susceptible cowpea cultivar used in the field for traps. After experiments, cowpea curculios were removed and placed in the freezer in preparation for dissections.

Results

This study evaluated insect damage on different genotypes in a no choice experiment to assess insect preference in a controlled environment providing only a single option for cowpea curculio feeding and oviposition activities. No significant differences in mean percentage of adult female cowpea curculios were observed among the genotypes (F=0.66; df=5, 25; $p < .660$). Mean percentages of females ranged from 34.6%-45.6% (Figure 3.3).

There were significant differences in the mean percentage of damaged peas across genotypes (F=5.73; df=5, 25; $p < .001$). All the genotypes, but one had significantly fewer damaged peas than the susceptible check. Within the group of genotypes, PI 533009, PI 533010, and PI 663152 had significantly fewer damaged peas than PI 582468, in addition to the susceptible check. The mean percentage of damaged peas of PI 582468 and the susceptible check were not different from each other (Table 3.3). The means ranged from 5.1% to 38.7% damaged peas. The susceptible check had the greatest damage of 38.7% and the genotype, PI 533010, had the lowest mean of 5.1% damaged peas (Figure 3.4). No significant differences in the mean number of stings on the pods were detected among the genotypes (F=1.01; df=5, 25; $p < .431$) (Table 3.3). The mean number of stings on the pods ranged from 6.81-13.21 (Figure 3.5).

Significant differences in the mean pod weight were observed among the genotypes (F=10.08; df=5, 25; $p < .0001$). PI 533009 had a significantly higher pod weight than other genotypes, with 14.4 g (Table 3.3). Significant differences in mean pea weight were observed among the genotypes (F=6.64; df=5, 25; $p < .0005$). Again, PI 533009 had a significantly higher mean pea weight than all of the genotypes, but one. PI 533009 was not significantly different from PI 582468 (Table 3.3).

Discussion

The purpose of a no choice test aids in determining insect preference by providing a single option for feeding and oviposition and eliminates variables present in an open choice field test that may influence results. However, chemical compounds and/or morphological barriers involved are not identified. The presence of sting marks on all genotypes indicates cowpea curculios would unitize the available source for food and reproduction. Both adult cowpea curculio sexes will feed on pods, but to examine oviposition preference, adult females needed to be identified. No external

features can be used for sex determination in cowpea curculios and dissections aided in determining average percentage of female cowpea curculios per genotype. Each of the genotypes had similar adult male to female ratios, as evidence of no statistical differences among means. Ideally, this means all genotypes had equal opportunity of feeding damage by both adult sexes and oviposition damage by adult females.

When presenting the cowpea curculios with a single option, it is evident that insects prefer the susceptible genotype. The least preferred genotype was PI 533010 with the lowest number of damaged peas followed by PI 533009 and PI 633152 with the second lowest percentage of pea damage. The lower percentages of damaged peas of PI 533009 and PI 533010 were not observed in the open choice field test (Chapter 2), and results from the no choice experiment provide insight into resistance to the cowpea curculio in these genotypes. The no choice test provided additional evidence for PI 663152 showing resistance in both types of experiments. Additional experiments measuring parameters for antibiosis are needed to confirm the presence of this resistance mechanism in the genotypes PI 533009, PI 53310, and PI 663152.

Tables and Figures

Table 3.1. Genotypes selected for no choice experiment.

Treatment No.	Genotype ID (PI No.)	Other Names	Origin
3	PI 533009	TARS-36	Puerto Rico, United States
4	PI 533010	TARS-62	Puerto Rico, United States
6	PI 582349	CPI 30783	Myanmar
9	PI 582468	UCR 347; Mn 150	Minnesota, United States
13	PI 663152	CR 22-2-21	South Carolina, United States
15	PI 663013 (Susc.)	California Blackeye 5	California, United States

Table 3.2. ANOVA output for parameters measured in evaluations of cowpea curculio damage on genotypes from no choice experiment in 2023.

Parameters	DF(n), df(d)	*F*	Pr>F
Pod weight (g)	5, 25	10.08	<.0001
# stings on pods	5, 25	1.01	.431
Pea weight (g)	5, 25	6.64	.0005
% damaged peas	5, 25	5.73	.001
% adult females	5, 25	0.66	.660

Table 3.3. Means of parameters measured for evaluations of cowpea curculio damage on the different genotypes from no choice experiment in 2023.

Genotypes	Pod weight (g)	# stings on pods	Pea weight (g)	% damaged peas	% adult females
PI 533009	14.40a	12.75a	4.97a	10.18c	34.68a
PI 533010	10.66b	6.81a	2.83c	5.11c	45.57a
PI 582349	8.56bc	10.45a	3.67bc	16.82bc	36.28a
PI 582468	9.16bc	11.19a	4.31ab	30.95ab	45.68a
PI 663152	6.67c	13.21a	2.79c	14.09c	38.75a
PI 663013 (Susc.)	7.2bc	11.21a	3.09c	38.69a	42.71a
LSD	2.57	6.58	0.998	15.67	17.03

*Means within columns followed by the same letter are not significantly different (P<0.05).

Figure 3.1. Insect rearing bag secured around two intact mature green pods and four adult cowpea curculios in no choice experiment.

Figure 3.2. Modified Tedders trap used for insect rearing (Riley et. al., 2015).

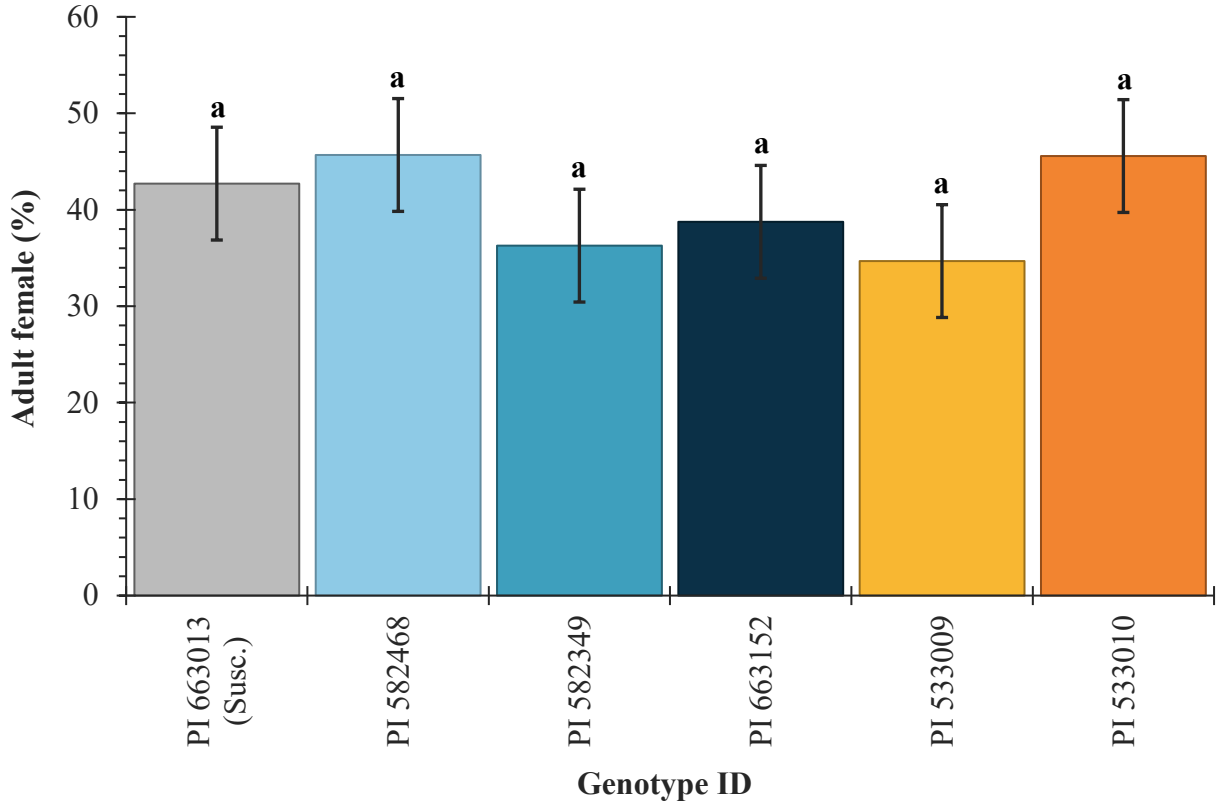

Figure 3.3. Bar plot showing mean percentage of adult female cowpea curculios on different genotypes in a no choice experiment after 72 h. Means followed by the same letter are not significantly different (P<0.05).

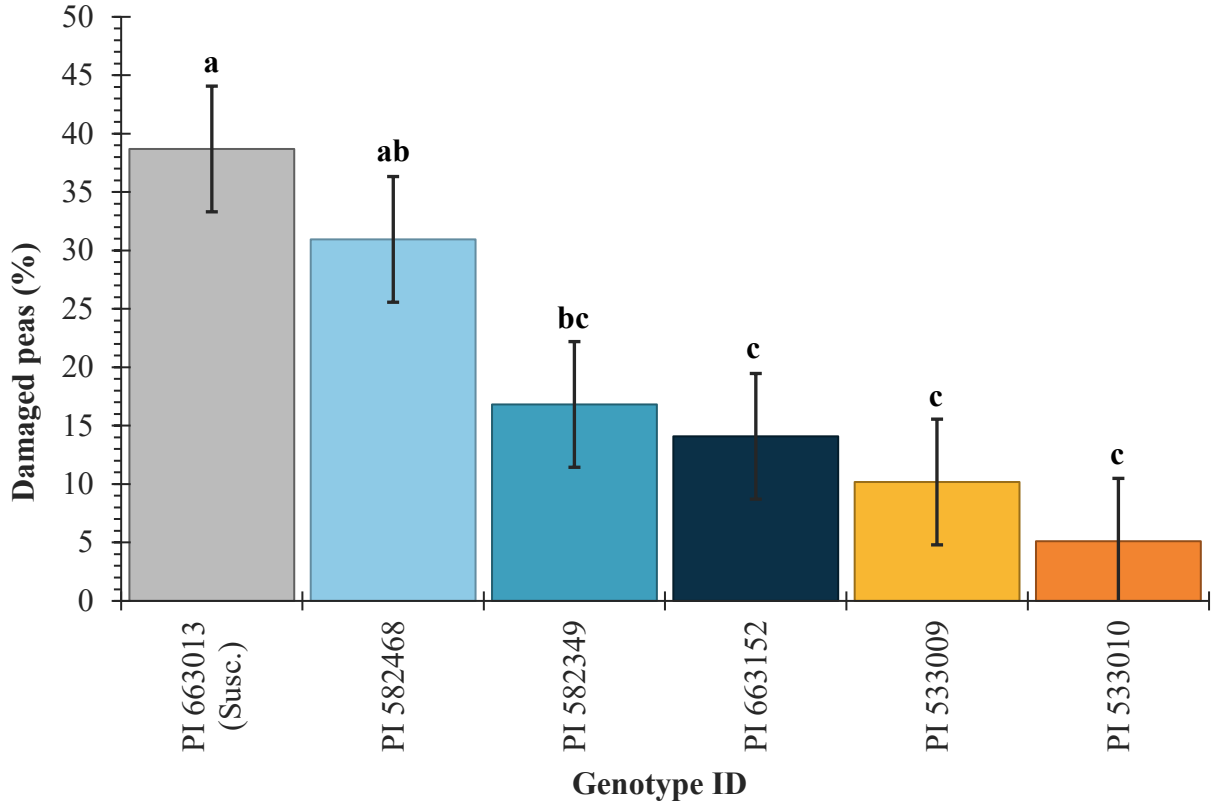

Figure 3.4. Bar plot showing mean percentage of damaged peas on different genotypes in a no choice experiment after 72 h. Means followed by the same letter are not significantly different ($P<0.05$).

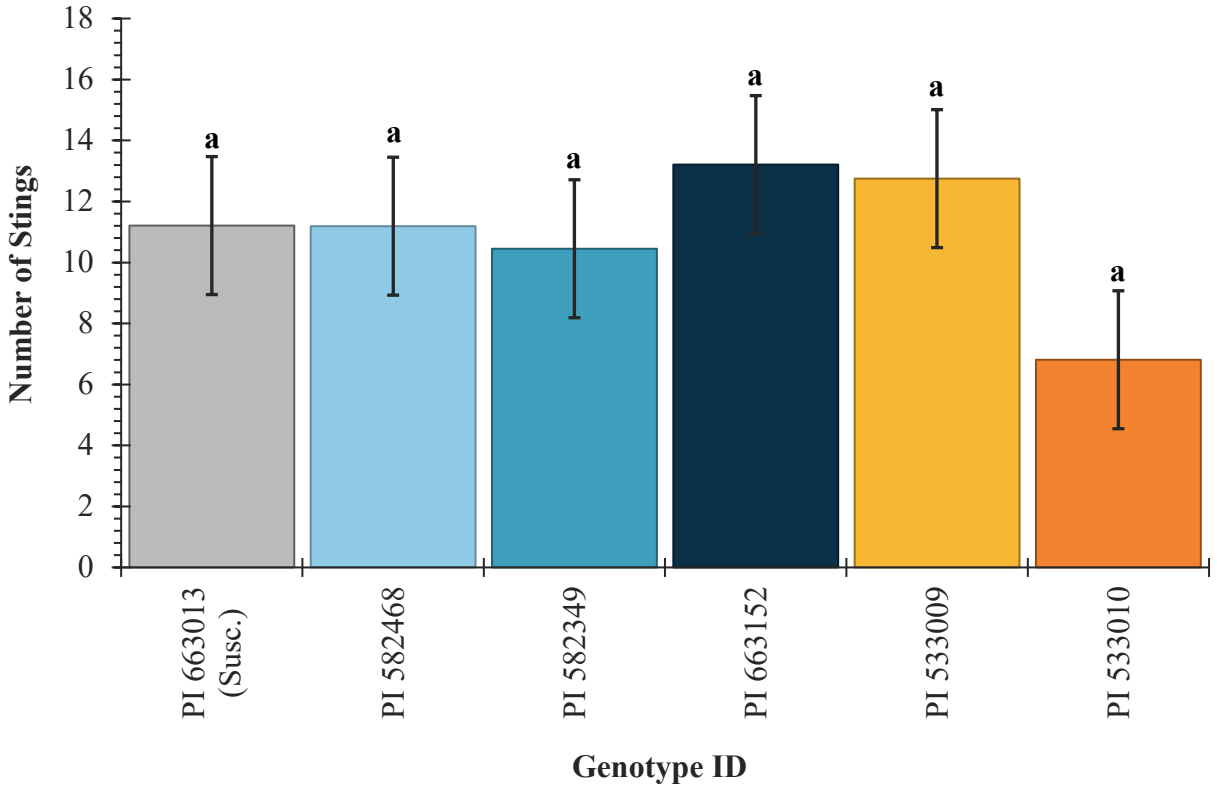

Figure 3.5. Bar plot showing mean number of sting marks on pods on different genotypes in a no choice experiment after 72 h. Means followed by the same letter are not significantly different ($P<0.05$).

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CHAPTER 4

SUMMARY

The goal of this study was to investigate host plant resistance in cowpea and identify germplasm with resistance to the cowpea curculio when cultivated in the Coastal Plain of Georgia. To accomplish this, several open choice experiments and a no choice experiment were conducted to evaluate cowpea curculio damage on different genotypes. The open choice experiments provided a natural field environment, giving insight into the cowpea curculio's host preference and overall genotype resistance. The no choice experiment provided a controlled environment to evaluate resistance by isolating insects to one genotype. Several genotypes showed potential resistance to the cowpea curculio, including PI 533009, PI 533010, PI 533011, PI 582448, PI 663152, and PI 664515. Both types of experiments showed that no genotype had complete resistance to the cowpea curculio, but many consistently had less damage than the susceptible check, suggesting the presence of some level of resistance. Specific factors involved with resistance mechanisms were not identified, however, the amount of pod and pea damage indicates possible factors of chemical or physical components contributing to a resistance mechanism. In the open choice experiments, the genotypes displayed a range of resistance reactions with varying levels of pod and seed damage among tests. The lower percentages of pod and seed damage suggests resistance is expressed primarily through antixenosis and antibiosis; however, the presence of tolerance cannot be dismissed. The no choice experiment confirmed the cowpea curculios preferred the susceptible check over the other genotypes when given a single option;

however, the cowpea curculios still utilized each genotype, but damage was significantly less than the susceptible.

Despite identifying genotypes with resistance to the cowpea curculio, these experiments still have limitations. The open choice experiments proved to be challenging with the cowpea genotypes varying in growth and morphology in addition to the variable environmental conditions that can influence each genotype differently. While the no choice experiment provided a controlled environment, other challenges of maintaining plants in an enclosed space such as a greenhouse or screenhouse created other insect issues. Additional experiments involving assessments of pod and seed secondary metabolites, pod wall thickness/strength, and larval mortality and survival need to be conducted to identify specific factors that may be influencing antixenosis and antibiosis resistance in the genotypes.

This study highlights the importance of finding the source(s) of resistance to the cowpea curculio and the need for resistant cultivars. Breeding for cowpea curculio resistance involves a complex approach, including the evaluation of germplasm and the identification and pyramiding of various resistance mechanisms to develop cultivars with improved resistance. Future breeding efforts should continue to build on the findings of this research and to combine multiple resistance mechanisms to develop cowpea curculio resistant cultivars. Additionally, utilizing the genetic and genomic resources that are becoming available would help to develop cowpea curculio resistant cultivars precisely and efficiently.

APPENDIX A

Information from GRIN on the accessions used in this study.

Treatment Number	Genotype ID (PI Number)	Other Names	Resistance	Agronomic traits
1	PI 142779	CUARENTANA; TVu 2260	Resistant to cowpea mosaic virus; Included in highest ranked entries for resistance to cowpea curculio (Chambliss, 1997)	Maturity (40-87 days), erect growth habit
2	PI 214354	TVu 1570	Resistant to aphid; Included in highest ranked entries for resistance to cowpea curculio (Chambliss, 1997)	Early maturity (40-60 days), semi-erect growth habit
3	PI 533009	TARS-36	Resistant to powdery mildew, cowpea mosaic, cercospora leafspot, root rots, cowpea curculio and cowpea weevil; Used as a susceptible check in evaluations for resistance to the cowpea curculio (Chambliss, 1997)	Daylength sensitive, late maturity (115-120 days), erect growth habit with medium long lateral vines
4	PI 533010	TARS-62	Resistant to powdery mildew, cowpea mosaic, cercospora leafspot, root rots and cowpea curculio	Daylength sensitive, late maturity (115-120 days), erect growth habit with medium long lateral vines
5	PI 533011	TARS-48	Resistant to powdery mildew, cowpea mosaic, cercospora leafspot, root rots, and cowpea curculio	Daylength sensitive, late maturity (115-120 days), erect growth habit with medium long lateral vines
6	PI 582349	CPI 30783	Included in highest ranked entries for resistance to cowpea curculio (Chambliss, 1997)	Early maturity (65-76 days), erect growth habit
7	PI 582393	IT82E-18	Broadly adapted, resistant to root-knot nematode	High yielding under drought in Mozambique; early maturity (65-75 days), photoperiod insensitive (Huynh et al., 2019)

Treatment Number	Genotype ID (PI Number)	Other Names	Resistance	Agronomic traits
8	PI 582448	UCR 306; IT82D-713	Included in highest ranked entries for resistance to cowpea curculio (Chambliss, 1997)	Medium maturity (75-100 days), erect growth habit
9	PI 582468	UCR 347; Mn 150	Tolerant to bacterial blight and drought; Included in highest ranked entries for resistance to cowpea curculio (Chambliss, 1997)	Extra-early maturity (60 days), erect growth habit (Davis et al., 1986)
10	PI 582512	IT84S-2049	Resistant to aphid, bacterial blight, viruses, root-knot nematode	High yielding under drought in Burkina Faso; medium maturity (75-100 days), erect growth habit (Huynh et al., 2019)
11	PI 582519	IT84S-2246-4	Resistant to scab, root knot nematode, aphid, bacterial blight, viruses, bruchid tolerant	High yielding under drought in Burkina Faso and Mozambique; early maturity (65-75 days), erect growth habit (Huynh et al., 2019)
12	PI 598343	IT89KD-288	Resistant to cercospora, root-knot nematode, aphid, thrips, bruchid	High yielding under drought in Burkina Faso and Nigeria; photoperiod sensitive, late maturity (85-120 days), semi-erect growth habit (Huynh et al., 2019)
13	PI 663152	CR 22-2-21	Relatively high level of antixenosis (non-preference) resistance to cowpea curculio	Medium maturity (90 days), erect growth habit (Cuthbert and Fery, 1976)
13	PI 664515	Bettergreen	Resistant to cowpea curculio, cercospora leaf spot, southern blight, rust, and powdery mildew	Homozygous for green cotyledon; early maturity (70 days), erect growth habit (Fery et al., 1993)

APPENDIX B

ANOVA output results and means of parameters measured in evaluations of genotypes from open choice experiment in 2021.

Genotypes	Clean pod wt (g)	Damaged pod wt (g)	% damaged pods	Clean pea wt (g)	Damaged pea wt (g)	% damaged peas	# of larvae
PI 142779	3.33d	190.67d	96.67ab	1.00e	81.67ef	60.33ab	6.67bc
PI 214354	45.67bc	340.33bcd	85.33cd	15.33cd	151.67bcd	13.33ef	0.33c
PI 533010	22.33bcd	460.67ab	95.00ab	4.33de	140.33bcde	7.00f	0.00c
PI 533011	17.0cd	372.00bc	91.33abc	1.67e	102.33def	55.00abc	5.33bc
PI 582349	87.67a	431.33ab	82.33d	29.33a	179.33ab	18.00ef	2.67c
PI 582393	4.0d	258.67cd	99.00a	0.667e	107.00cdef	58.00ab	12.00ab
PI 582448	24.67bcd	354.33bc	93.67abc	9.00cde	175.00ab	24.33def	3.67c
PI 582468	44.33bc	326.0bcd	85.67cd	17.33bc	186.33ab	32.33cde	19.00a
PI 582512	54.33b	445.0ab	88.00bcd	18.00bc	172.33abc	43.67bcd	5.67bc
PI 582519	33.0bcd	429.67ab	92.33abc	10.00cde	224.00a	23.33def	0.00c
PI 598343	44.67bc	344.33bcd	88.00bcd	9.00cde	72.67f	6.67f	0.00c
PI 663152	17.67cd	267.00cd	93.33abc	27.00ab	150.67bcd	13.00ef	2.00c
GMO Cowpea AAI	7.67d	553.0a	98.00a	0.867e	157.67abcd	70.33a	0.00c
Snap bean	-	-	-	-	-	-	-
DF(n), df(d)	12, 24	12, 24	12, 24	12, 24	12, 24	12, 24	12, 24
F	4.63	3.36	3.09	6.69	3.66	7.33	5.31
Pr>F	.0007	.005	.009	<.0001	.003	<.0001	.0003
LSD	32.45	154.72	8.66	11.04	67.52	23.93	7.14

*Means within columns followed by the same letter are not significantly different (P<0.05).

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